Probability

MATH 356 · Scribed Course Notes · Module 2

Definitions
Examples
Results
Notes

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Note. This module continues Module 1 (Sections 1–10, manuscript pages 1–49). The conventions in Notes 1–4 there apply.

11 The Central Limit Theorem (Part 1)

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Let $(X_k)_{k\geq 1}$ be independent $\mathrm{Ber}(p)$-distributed random variables. We have shown that $S_n = \sum_{i=1}^n X_i$ is $\mathrm{Bin}(n,p)$-distributed. Consequently, $\mathbb{E}S_n = np$, and $\mathrm{Var}(S_n) = np(1-p)$ by the computation we did with the $\mathrm{Bin}(n,p)$ distribution.

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Now define

$$S_n^* := \frac{S_n - np}{\sqrt{np(1-p)}},$$

the *standardization* of S_n. Then $\mathbb{E}S_n^* = 0$, and $\mathrm{Var}(S_n^*) = 1$.

Theorem

For $p \in (0,1)$, if $S_n \sim \mathrm{Bin}(n,p)$, then $S_n^* := (S_n - np)/\sqrt{np(1-p)}$ satisfies that $\forall a < b \in \mathbb{R}$,

$$\mathbb{P}\big(S_n^* \in [a,b]\big) \xrightarrow{n\to\infty} \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{x^2}{2}}\, dx = \Phi(b) - \Phi(a).$$

Remark

This holds even if $p = p(n)$ depends on n, provided that $np(n) \to \infty$ and $n(1-p(n)) \to \infty$. □

Before stating the proof, we introduce examples and the necessary lemmas.

Example

If $p(n) = \dfrac{1}{\sqrt{n}}$, then

$$\mathbb{P}\left(\frac{S_n - \sqrt{n}}{\sqrt{n \cdot \frac{1}{\sqrt{n}} \cdot \left(1 - \frac{1}{\sqrt{n}}\right)}} \in [a,b]\right) \xrightarrow{n\to\infty} \Phi(b) - \Phi(a).$$

The typical fluctuation / dispersion / spread of S_n for this example is of order $\sqrt{np(n)}$, since $1 - \frac{1}{\sqrt{n}} \to 1$ as $n \to \infty$. □

Since $\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}$, with $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$, it becomes necessary to study $n!$ and $k!$ to understand $\binom{n}{k}$.

11.1 Stirling-Type Estimates

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Input. $\dfrac{d}{dx}\, x\log x = \log x + 1, \quad$ so $\quad \dfrac{d}{dx}\big(x\log x - x\big) = \log x.$

Proposition

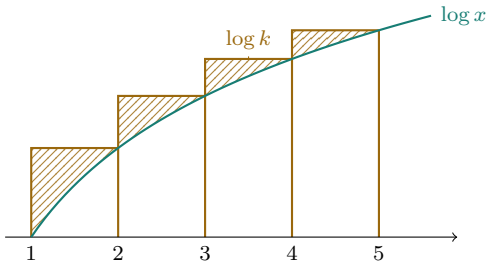
For $n \geq 2$,

$$\big|\log(n!) - (n\log n - n + 1)\big| \;\leq\; \log n, \qquad \text{equivalently,} \qquad \frac{1}{n} \;\leq\; n! \cdot \left[e\left(\frac{n}{e}\right)^n\right]^{-1} \;\leq\; n.$$

Proof. Since log is increasing, $\log k \geq \int_{k-1}^{k} \log x\,dx$ for every $k \geq 2$; summing over $k = 2, \dots, n$,

$$\log(n!) = \sum_{k=1}^{n} \log k \;\geq\; \int_{1}^{n} \log x\,dx = \Big[x\log x - x\Big]_{1}^{n} = n\log n - n + 1.$$

(The sum is a right-endpoint Riemann sum of an increasing function, so it lies *above* the integral — exactly what the staircase shows.)



In the other direction, $\log k \leq \int_{k}^{k+1} \log x\,dx$; summing over $k = 1, \dots, n-1$,

$$\log(n!) - \log n = \log\big((n-1)!\big) = \sum_{k=1}^{n-1} \log k \;\leq\; \int_{1}^{n} \log x\,dx = n\log n - n + 1.$$

Together,

$$0 \;\leq\; \log(n!) - (n\log n - n + 1) \;\leq\; \log n,$$

which gives both displays of the proposition (the second, in fact, with 1 in place of $\frac{1}{n}$). □

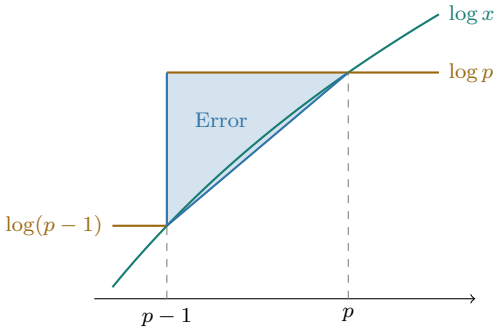
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Theorem (Stirling's Formula)

$$\lim_{n\to\infty} n! \cdot \left[\sqrt{2\pi n}\left(\frac{n}{e}\right)^{n}\right]^{-1} = 1.$$

Proof idea. Really, for each $p \geq 2$ the gap between $\log p$ and the integral over $[p-1, p]$ is about half the increment of log — the area of the little triangle between the chord and the curve:

$$\log p - \int_{p-1}^{p} \log x\,dx \;\approx\; \tfrac{1}{2}\big(\log p - \log(p-1)\big).$$



So

$$\begin{aligned}\log(n!) = \sum_{k=1}^{n} \log k \;&\approx\; \int_{1}^{n} \log x\,dx + \sum_{k=2}^{n} \tfrac{1}{2}\big(\log k - \log(k-1)\big) \\ &= n\log n - n + 1 + \tfrac{1}{2}\log n,\end{aligned}$$

and therefore

$$n! \;\approx\; \exp\Big(n\log n - n + 1 + \tfrac{1}{2}\log\big(n\big)\Big) = \Big(\frac{n}{e}\Big)^n \cdot e \cdot \sqrt{n}\,.$$

Being more careful about the error terms gives the result: the careful analysis replaces the heuristic constant $e \approx 2.72$ in front of $\sqrt{n}$ by the true $\sqrt{2\pi} \approx 2.51$.

11.2 Asymptotic Notation

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The proof of the theorem occupies the rest of this section. Notation.

Definition (asymptotic notation)

For functions f, g of n:

- $f(n) \sim g(n)$ as $n \to \infty$ means $\dfrac{f(n)}{g(n)} \to 1$ as $n \to \infty$;
- $f(n) = o(g(n))$ as $n \to \infty$ means $\dfrac{|f(n)|}{g(n)} \to 0$ as $n \to \infty$;
- $f(n) = o(1)$ means $|f(n)| \to 0$ as $n \to \infty$;
- $f(n) = (1 + o(1))\, g(n)$ as $n \to \infty$ means $\Big|\dfrac{f(n)}{g(n)} - 1\Big| \to 0$, i.e. $\dfrac{f(n)}{g(n)} \to 1$, i.e. $f(n) \sim g(n)$.

□

Reminders.

- Write $q := 1 - p$, so $np(1-p) = npq$.
- Stirling: $n! \sim \sqrt{2\pi n}\,\big(\frac{n}{e}\big)^n$.
- Taylor expansion: $\log(1+x) = x - \dfrac{x^2}{2} + o(x^2)$ as $x \to 0$.
- Recall also $\liminf_{n\to\infty} f(n)$ (used for the law of large numbers below).

11.3 The Local Limit Theorem

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Lemma

For any $a < b \in \mathbb{R}$: if $k = k(n)$ is such that $\frac{k-np}{\sqrt{npq}} \in [a, b]$ for all n, then as $n \to \infty$,

$$\mathbb{P}(S_n = k) = (1 + o(1)) \cdot \frac{1}{\sqrt{2\pi npq}} \exp\Big(-\frac{(k-np)^2}{2npq}\Big) = (1 + o(1)) \cdot \frac{1}{\sqrt{npq}}\, \varphi\Big(\frac{k-np}{\sqrt{npq}}\Big).$$

Proof. Throughout, $k - np \in [a\sqrt{npq},\, b\sqrt{npq}]$, so $k \sim np \to \infty$ and $n - k \sim nq \to \infty$, and Stirling applies to all three factorials:

$$\begin{aligned}\mathbb{P}(S_n = k) = \frac{n!}{k!\,(n-k)!}\, p^k q^{n-k} &\sim \frac{\sqrt{2\pi n}\,\big(\frac{n}{e}\big)^n}{\sqrt{2\pi k}\,\big(\frac{k}{e}\big)^k \cdot \sqrt{2\pi(n-k)}\,\big(\frac{n-k}{e}\big)^{n-k}}\, p^k q^{n-k} \\ &= \frac{\sqrt{2\pi n}}{\sqrt{2\pi k}\,\sqrt{2\pi(n-k)}} \Big(\frac{np}{k}\Big)^k \Big(\frac{nq}{n-k}\Big)^{n-k} \sim \frac{1}{\sqrt{2\pi npq}} \Big(\frac{np}{k}\Big)^k \Big(\frac{nq}{n-k}\Big)^{n-k},\end{aligned}$$

using $k \sim np$ and $n-k \sim nq$ in the square-root factors. It remains to show that the last product is $(1+o(1))\exp\big(-\frac{(k-np)^2}{2npq}\big)$. To do so, write $d := k - np$ and

$$\frac{np}{k} = 1 - \frac{d}{k}, \qquad \frac{nq}{n-k} = 1 + \frac{d}{n-k},$$

so that, by the Taylor expansion of $\log(1+x)$ (here $x = O(1/\sqrt{npq}\,)\cdot O(1) \to 0$),

$$\begin{aligned}
\Big(\frac{np}{k}\Big)^k \Big(\frac{nq}{n-k}\Big)^{n-k} &= \exp\Big(k\log\frac{np}{k} + (n-k)\log\frac{nq}{n-k}\Big) \\
&= \exp\Big(k\Big(-\frac{d}{k} - \frac{d^2}{2k^2}\Big) + (n-k)\Big(\frac{d}{n-k} - \frac{d^2}{2(n-k)^2}\Big) + o(1)\Big) \\
&= \exp\Big(-\frac{d^2}{2k} - \frac{d^2}{2(n-k)} + o(1)\Big) = \exp\Big(-\frac{q\,d^2}{2kq} - \frac{p\,d^2}{2(n-k)p} + o(1)\Big) \\
&\sim \exp\Big(-\frac{d^2}{2npq}\Big),
\end{aligned}$$

since $kq \sim npq$ and $(n-k)p \sim nqp$, while $d^2 = O(npq)$ keeps every error term $o(1)$. □

11.4 Proof of the Central Limit Theorem

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Proof of the theorem.

$$\mathbb{P}\big(S_n^* \in [a,b]\big) = \mathbb{P}\Big(\frac{S_n - np}{\sqrt{npq}} \in [a,b]\Big) = \mathbb{P}\big(np + a\sqrt{npq} \;\le\; S_n \;\le\; np + b\sqrt{npq}\big) = \sum_{k=\lceil np + a\sqrt{npq}\rceil}^{\lfloor np + b\sqrt{npq}\rfloor} \mathbb{P}(S_n = k).$$

By the lemma (applied uniformly over this range of k),

$$\mathbb{P}\big(S_n^* \in [a,b]\big) \sim \sum_{k=\lceil np + a\sqrt{npq}\rceil}^{\lfloor np + b\sqrt{npq}\rfloor} \frac{1}{\sqrt{2\pi npq}} \exp\Big(-\frac{(k-np)^2}{2npq}\Big).$$

Write $x_k = \dfrac{k - np}{\sqrt{npq}}$; then this becomes

$$\sum_{a \le x_k \le b} \frac{1}{\sqrt{npq}} \cdot \frac{1}{\sqrt{2\pi}} e^{-x_k^2/2} = \sum_{a \le x_k \le b} \frac{1}{\sqrt{npq}} \cdot \varphi(x_k) \xrightarrow{\;n\to\infty\;} \int_a^b \varphi(x)\,dx,$$

a Riemann approximation: the points x_k run through $[a,b]$ with mesh $\frac{1}{\sqrt{npq}} \to 0$. □

11.5 The Law of Large Numbers

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Corollary ((Binomial) Law of Large Numbers)

Let $(X_i)_{i\ge 1}$ be iid $\mathrm{Ber}(p)$, and $S_n = X_1 + \cdots + X_n$. Then for all $\varepsilon > 0$,

$$\mathbb{P}\big(|S_n/n - p| > \varepsilon\big) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Proof. Fix any $\varepsilon > 0$ (small) and $C > 0$ (large). Then for n large, $C\sqrt{npq} < \varepsilon n$, so

$$\mathbb{P}\big(|S_n/n - p| < \varepsilon\big) = \mathbb{P}\big(|S_n - np| < \varepsilon n\big) \;\geq\; \mathbb{P}\big(|S_n - np| < C\sqrt{npq}\big).$$

Thus, by the central limit theorem applied to the right-hand side,

$$\liminf_{n\to\infty} \mathbb{P}\big(|S_n/n - p| < \varepsilon\big) \;\geq\; \liminf_{n\to\infty} \mathbb{P}\big(|S_n - np| < C\sqrt{npq}\big) = \int_{-C}^{C} \varphi(x)\,dx.$$

This holds for all C, so

$$\liminf_{n\to\infty} \mathbb{P}\big(|S_n/n - p| < \varepsilon\big) \;\geq\; \sup_{C>0} \int_{-C}^{C} \varphi(x)\,dx = \int_{-\infty}^{\infty} \varphi(x)\,dx = 1.$$

But also $\mathbb{P}\big(|S_n/n - p| < \varepsilon\big) \leq 1$ for every n, so $\mathbb{P}\big(|S_n/n - p| < \varepsilon\big) \to 1$, i.e. $\mathbb{P}\big(|S_n/n - p| > \varepsilon\big) \to 0$. □

11.6 Confidence Intervals

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Given a coin with *unknown* probability p of heads, flip the coin n times and let $S_n = \#\text{heads}$. Then $\hat{p} = S_n/n$ is the *empirical* average, and

$$\begin{aligned}\mathbb{P}\big(|\hat{p} - p| \leq \varepsilon\big) = \mathbb{P}\big(|S_n - np| \leq \varepsilon n\big) &= \mathbb{P}\Big(\frac{|S_n - np|}{\sqrt{npq}} \leq \frac{\varepsilon}{\sqrt{pq}}\sqrt{n}\Big) \\ &\approx \Phi\Big(\frac{\varepsilon}{\sqrt{pq}}\sqrt{n}\Big) - \Phi\Big(-\frac{\varepsilon}{\sqrt{pq}}\sqrt{n}\Big) = 2\,\Phi\Big(\frac{\varepsilon}{\sqrt{pq}}\sqrt{n}\Big) - 1.\end{aligned}$$

Berry–Esseen theorem: the error in "≈" here is at most $C/\sqrt{npq}$, for an absolute constant C. Now $\frac{\varepsilon}{\sqrt{pq}} = \frac{\varepsilon}{\sqrt{p(1-p)}}$, and $p(1-p) \leq \frac{1}{4}$, so $\frac{\varepsilon}{\sqrt{p(1-p)}} \geq 2\varepsilon$. Since Φ is increasing,

$$\mathbb{P}\big(|\hat{p} - p| \leq \varepsilon\big) \;\gtrsim\; 2\,\Phi\big(2\varepsilon\sqrt{n}\big) - 1$$

("≥", up to the error of the normal approximation), whatever the unknown p is. If we want $\mathbb{P}(|\hat{p} - p| \leq 0.05) \geq 0.99$, then we need $2\Phi(0.1\sqrt{n}) - 1 \geq 0.99$, i.e. $\Phi(0.1\sqrt{n}) \geq 0.995$, i.e. $0.1\sqrt{n} \geq z_{0.995} = 2.5758\ldots$, i.e. $n \geq 664$.

Note

Two parameters: ε (the error) and 0.99 (the confidence). Polling ← typical confidence 9/10. □

11.7 The Maximum Likelihood Estimator

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If $S_n = k$, what is the *most likely* value of p?

$$p^* = \arg\max\left(\binom{n}{k} p^k (1-p)^{n-k} :\; 0 \leq p \leq 1\right) ?$$

It is $\dfrac{k}{n}$:

$$\begin{aligned}\frac{d}{dp}\big(p^k(1-p)^{n-k}\big) &= kp^{k-1}(1-p)^{n-k} - (n-k)p^k(1-p)^{n-k-1} \\ &= p^{k-1}(1-p)^{n-k-1}\big(k(1-p) - (n-k)p\big).\end{aligned}$$

Set $k(1-p) = (n-k)p$, i.e. $k = np$. (For $0 < k < n$ the derivative is positive to the left of $p = k/n$ and negative to the right, so this critical point is the maximum; for $k = 0$ or $k = n$ the maximum sits at the endpoint $p = 0$ or $p = 1$, which is again k/n.)

11.8 Sampling With vs. Without Replacement

Recall the hypergeometric PMF: $X \sim \mathrm{Hyp}(N, m, k)$ (N = urn size, m = #red, k = #draws) if

$$\mathbb{P}(X = j) = \frac{\binom{m}{j}\binom{N-m}{k-j}}{\binom{N}{k}} \qquad \text{for } 0 \le j \le k,$$

$= \mathbb{P}(j$ out of k red when sampling *without* repetition from m red, $N - m$ blue). Rewrite as

$$\frac{\frac{(m)_j}{j!} \cdot \frac{(N-m)_{k-j}}{(k-j)!}}{\frac{(N)_k}{k!}} = \binom{k}{j} \frac{(m)_j\,(N-m)_{k-j}}{(N)_k}.$$

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If j, k are fixed and $N, m \to \infty$ such that $\frac{m}{N} \to p$, then

$$\frac{(m)_j\,(N-m)_{k-j}}{(N)_k} = \underbrace{\frac{m(m-1)\cdots(m-j+1)}{N(N-1)\cdots(N-j+1)}}_{\to\, p^j} \cdot \underbrace{\frac{(N-m)(N-m-1)\cdots(N-m-(k-j)+1)}{(N-j)(N-j-1)\cdots(N-j-(k-j)+1)}}_{\to\, (1-p)^{k-j}}.$$

So

$$\mathbb{P}(X = j) \longrightarrow \binom{k}{j} p^j (1-p)^{k-j}.$$

(Justifies sampling without replacement: for a large population, sampling k without replacement is approximately $\mathrm{Bin}(k, p)$, i.e. the same as sampling with replacement.)

11.9 Simple Random Walk

Let the steps be independent, $+1$ with probability p and -1 with probability $1 - p$, and let S_{2n} be the position after $2n$ steps. Then $S_{2n} = 2k$ exactly when $n + k$ of the $2n$ steps are $+1$, so

$$\mathbb{P}(S_{2n} = 2k) = \binom{2n}{n+k} p^{n+k}(1-p)^{n-k} = \mathbb{P}\big(\mathrm{Bin}(2n, p) = n + k\big).$$

For $p = \frac{1}{2}$,

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n}\frac{1}{2^{2n}} \sim \frac{1}{\sqrt{\pi n}}, \quad \text{since by Stirling} \quad \frac{(2n)!}{n!\,n!} = (1+o(1))\,\frac{\sqrt{2\pi(2n)}\left(\frac{2n}{e}\right)^{2n}}{\left[\sqrt{2\pi n}\left(\frac{n}{e}\right)^{n}\right]^2} = (1+o(1))\,\frac{2^{2n}}{\sqrt{\pi n}}.$$

By the local limit theorem (with $2n$ trials and $p = q = \frac{1}{2}$, so that "npq" is $n/2$),

$$\mathbb{P}(S_{2n} = 2k) \sim \frac{1}{\sqrt{\pi n}} \exp\Big(-\frac{k^2}{n}\Big) \qquad \text{for } k = O(\sqrt{n}).$$

Further topics. Transience vs. recurrence; law of large numbers; $p \neq \frac{1}{2}$.

12 The Poisson Approximation

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Definition (Poisson distribution)

$X \sim \mathrm{Poi}(\lambda)$ if $\mathbb{P}(X = k) = \dfrac{\lambda^k}{k!} e^{-\lambda}$ for $k \geq 0$. Why is it a PMF?[a] □

[a] Because $\sum_{k \geq 0} \frac{\lambda^k}{k!} = e^{\lambda}$, so the weights are non-negative and sum to 1.

Let $S_n \sim \mathrm{Bin}(n, \lambda/n)$. Then, for fixed k, as $n \to \infty$,

$$\mathbb{P}(S_n = k) = \frac{n!}{k!\,(n-k)!} \Big(\frac{\lambda}{n}\Big)^k \Big(1 - \frac{\lambda}{n}\Big)^{n-k} = \frac{n(n-1)\cdots(n-k+1)}{k!} \frac{\lambda^k}{n^k} \Big(1 - \frac{\lambda}{n}\Big)^n \Big(1 - \frac{\lambda}{n}\Big)^{-k}$$
$$= \frac{\lambda^k}{k!} \Big(1 - \frac{\lambda}{n}\Big)^n \cdot \underbrace{\frac{n(n-1)\cdots(n-k+1)}{n \cdot n \cdots n} \cdot \Big(1 - \frac{\lambda}{n}\Big)^{-k}}_{\to 1} \longrightarrow \frac{\lambda^k}{k!} e^{-\lambda}.$$

Proposition (moments of the Poisson distribution)

If $X \sim \mathrm{Poi}(\lambda)$ then $\mathbb{E}X = \lambda$ and $\mathbb{E}[X(X-1)] = \lambda^2$; hence $\mathbb{E}[X^2] = \lambda^2 + \lambda$ and

$$\mathrm{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

Proof.

$$\mathbb{E}X = \sum_{k \geq 0} k \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k \geq 1} \frac{\lambda^k}{(k-1)!} e^{-\lambda} = \lambda \sum_{k \geq 0} \frac{\lambda^k}{k!} e^{-\lambda} = \lambda,$$
$$\mathbb{E}[X(X-1)] = \sum_{k \geq 0} k(k-1) \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k \geq 2} \frac{\lambda^k}{(k-2)!} e^{-\lambda} = \lambda^2 \sum_{k \geq 0} \frac{\lambda^k}{k!} e^{-\lambda} = \lambda^2.$$ □

12.1 Normal versus Poisson

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We have seen:

- For $0 < p < 1$ constant, $\mathrm{Bin}(n, p) \approx np + \sqrt{npq} \cdot \mathcal{N}(0, 1)$.
- For $p = \lambda/n$, λ constant, $\mathrm{Bin}(n, p) \approx \mathrm{Poi}(np) = \mathrm{Poi}(\lambda)$.

Theorem (Berry–Esseen: quality of the normal approximation)

For every $a \in \mathbb{R}$,

$$\Big| \mathbb{P}\Big(\frac{S_n - np}{\sqrt{npq}} \leq a \Big) - \mathbb{P}\big(\mathcal{N}(0,1) \leq a\big) \Big| \;\leq\; \frac{3}{\sqrt{npq}}.$$

(The general Berry–Esseen theorem bounds the error for a sum of n iid variables by $C\,\mathbb{E}|X_1 - \mu|^3/(\sigma^3\sqrt{n})$ with an absolute constant $C < 0.48$; for $\mathrm{Ber}(p)$, $\mathbb{E}|X_1 - p|^3/\sigma^3 = (p^2 + q^2)/\sqrt{pq} \leq 1/\sqrt{pq}$, so the constant 3 above is safe.)

Fact (quality of the Poisson approximation)

For any $A \subseteq \{0, 1, 2, \dots\}$,

$$\big| \mathbb{P}(S_n \in A) - \mathbb{P}\big(\mathrm{Poi}(np) \in A\big) \big| \;\leq\; np^2.$$

When to use Normal vs. Poisson?

- Rule of thumb. If $np > 10$ (and, by the symmetry $p \leftrightarrow q$, $nq > 10$) then $\mathcal{N}(0,1)$ is pretty good. Improved by the "*continuity correction*": for $a, b \in \mathbb{Z}$,

$$\mathbb{P}(a \le S_n \le b) = \mathbb{P}\big(a - \tfrac{1}{2} \le S_n \le b + \tfrac{1}{2}\big) \approx \Phi\Big(\frac{b + \frac{1}{2} - np}{\sqrt{npq}}\Big) - \Phi\Big(\frac{a - \frac{1}{2} - np}{\sqrt{npq}}\Big).$$

- If np^2 is small then $\mathrm{Poi}(np)$ is pretty good.
- For $\frac{1}{n} \ll p \ll \frac{1}{\sqrt{n}}$, both are good.

12.2 Numbers

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$\mathrm{Bin}(10, \frac{1}{10}) \le 1$. Here $np = 1$, $np^2 = 0.1$, $npq = 0.9$:

$$\Big|\mathbb{P}\big(\mathrm{Bin}(10, \tfrac{1}{10}) \le 1\big) - \mathbb{P}\big(\mathrm{Poi}(1) \le 1\big)\Big| \le 0.001,$$
$$\Big|\mathbb{P}\big(\mathrm{Bin}(10, \tfrac{1}{10}) \le 1\big) - \mathbb{P}\Big(\mathcal{N}(0,1) \le \underbrace{\tfrac{1-np}{\sqrt{npq}}}_{=0}\Big)\Big| \approx 0.23.$$

(Exactly: $\mathbb{P}(\mathrm{Bin}(10, \frac{1}{10}) \le 1) = 0.9^{10} + 0.9^9 = 0.7361$, $\mathbb{P}(\mathrm{Poi}(1) \le 1) = 2e^{-1} = 0.7358$, difference 0.0003; and $\Phi(0) = \frac{1}{2}$, difference 0.236. The Poisson approximation wins by a mile here.)

$\mathrm{Bin}(40, \frac{1}{2}) = 20$. Here $np = 20$, $np^2 = 10$, $npq = 10$:

$$\Big|\mathbb{P}\big(\mathrm{Bin}(40, \tfrac{1}{2}) = 20\big) - \mathbb{P}\big(\mathrm{Poi}(20) = 20\big)\Big| \approx 0.04 \qquad \text{(about 30\% off)},$$
$$\Big|\mathbb{P}\big(\mathrm{Bin}(40, \tfrac{1}{2}) = 20\big) - \mathbb{P}\big(-\tfrac{1}{2} \le \mathcal{N}(0,10) \le \tfrac{1}{2}\big)\Big| \approx 0.0003 \qquad \text{(about 0.2\% off)},$$

where $\mathcal{N}(0,10)$ has variance $npq = 10$ and the second line is the continuity correction $\mathbb{P}(19.5 \le \mathrm{Bin}(40, \frac{1}{2}) \le 20.5)$. (Exactly: $\mathbb{P}(\mathrm{Bin}(40, \frac{1}{2}) = 20) = \binom{40}{20} 2^{-40} = 0.1254$, $\mathbb{P}(\mathrm{Poi}(20) = 20) = 20^{20} e^{-20}/20! = 0.0888$, difference 0.0365; and $\mathbb{P}(-\frac{1}{2} \le \mathcal{N}(0,10) \le \frac{1}{2}) = 2\Phi(0.5/\sqrt{10}) - 1 = 0.1256$, difference 0.0003. The normal approximation wins by a mile here.)

12.3 The Poisson Model

Heuristic. If X counts the number of *rare* events, after *many* approximately *independent* trials, then

$$X \approx \mathrm{Poi}(\lambda) \qquad \text{for } \lambda = \mathbb{E}X.$$

Example (factory)

In a factory which averages 3 accidents/month, $\mathbb{P}(2 \text{ accidents in October})$?
Modelling assumptions: individual workers unlikely to have an accident; accidents of different workers approximately independent of each other. Approximate independence is enough! So the number of accidents in October is $\approx \mathrm{Poi}(3)$, and

$$\mathbb{P}(2 \text{ accidents in October}) \approx \frac{3^2}{2!} e^{-3} = \tfrac{9}{2} e^{-3} \approx 0.224.$$

□

Example (hats)

$\mathbb{P}(k \text{ among } n \text{ get the correct hat}) \longrightarrow \dfrac{e^{-1}}{k!} = \mathbb{P}(\mathrm{Poi}(1) = k)$ as $n \to \infty$. So $N = \#\text{people getting the correct hat} \longrightarrow \mathrm{Poi}(1)$. (But if $X_i = \mathbf{1}\{i \text{ gets own hat}\}$ then $X_1, \dots, X_n$ are *not* independent — the heuristic only needs approximate independence.) □

13 The Exponential Distribution

Source manuscript page 63

Definition (exponential distribution)

A random variable X is $\mathrm{Exp}(\lambda)$-distributed if $\mathbb{P}(X > t) = e^{-\lambda t}$ for $t \geq 0$, i.e. X has CDF

$$F(t) = \begin{cases} 0, & \text{if } t < 0, \\ 1 - e^{-\lambda t}, & \text{if } t \geq 0. \end{cases}$$

$\Rightarrow X$ has density $f(t) = \lambda e^{-\lambda t} \mathbf{1}_{t \geq 0}$. □

Mean. Integrating by parts with $u = \lambda t$, $dv = e^{-\lambda t}\,dt$ (so $du = \lambda\,dt$, $v = -e^{-\lambda t}/\lambda$):

$$\mathbb{E}X = \int_0^\infty \lambda t e^{-\lambda t}\,dt = \big[uv\big]_0^\infty - \int_0^\infty v\,du = \big[-te^{-\lambda t}\big]_0^\infty + \int_0^\infty e^{-\lambda t}\,dt = 0 + \Big[-\frac{e^{-\lambda t}}{\lambda}\Big]_0^\infty = \frac{1}{\lambda}.$$

Variance. With $u = t^2$, $dv = \lambda e^{-\lambda t}\,dt$ (so $du = 2t\,dt$, $v = -e^{-\lambda t}$):

$$\mathbb{E}[X^2] = \int_0^\infty t^2 f(t)\,dt = \int_0^\infty t^2 \lambda e^{-\lambda t}\,dt = \big[-t^2 e^{-\lambda t}\big]_0^\infty + \int_0^\infty 2te^{-\lambda t}\,dt = 0 + \frac{2}{\lambda}\int_0^\infty \lambda t e^{-\lambda t}\,dt = \frac{2}{\lambda}\,\mathbb{E}X = \frac{2}{\lambda^2}.$$

So

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

13.1 Memorylessness

Source manuscript page 64

Proposition (exponentials are memoryless)

If $X \sim \mathrm{Exp}(\lambda)$ then for all $s, t \geq 0$,

$$\mathbb{P}(X > t + s \mid X > t) = \frac{\mathbb{P}(X > t+s)}{\mathbb{P}(X > t)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} = e^{-\lambda s} = \mathbb{P}(X > s).$$

E.g. radioactive decay $\rightsquigarrow \mathbb{P}(\text{decays in time } 1/n) = 1 - e^{-\lambda/n} \approx \lambda/n$.

Note

The geometric distribution has a *discrete* version of this: for integers $k, \ell \geq 0$,

$$\mathbb{P}(G > k + \ell \mid G > k) = \frac{(1-p)^{k+\ell}}{(1-p)^k} = (1-p)^\ell = \mathbb{P}(G > \ell).$$

□

13.2 The Exponential as a Limit of Geometrics

Fix λ, and let n be large enough that $\lambda/n < 1$. If $(X_i)_{i\geq 1}$ are independent $\mathrm{Ber}(\lambda/n)$ then $G = \min(i : X_i = 1)$ is $\mathrm{Geom}(\lambda/n)$. Then, for $t > 0$,

$$\mathbb{P}(G/n \geq t) = \mathbb{P}(G \geq tn) = \mathbb{P}\big(G \geq \lceil tn \rceil\big) = \mathbb{P}\big(G > \lceil tn \rceil - 1\big) = \Big(1 - \frac{\lambda}{n}\Big)^{\lceil tn \rceil - 1}$$
$$= \underbrace{\Big(1 - \frac{\lambda}{n}\Big)^{tn}}_{\to\, e^{-\lambda t}} \cdot \underbrace{\Big(1 - \frac{\lambda}{n}\Big)^{\lceil tn \rceil - tn - 1}}_{\to\, 1} \longrightarrow e^{-\lambda t} \qquad \text{as } n \to \infty$$

(the second exponent lies in $(-1, 0]$, so that factor is squeezed between 1 and $(1 - \lambda/n)^{-1} \to 1$). So $G/n \rightsquigarrow \mathrm{Exp}(\lambda)$ as $n \to \infty$.

14 Poisson Processes (Not Covered)

Note (instructor's remark)

"The notes continue on to discuss Poisson processes — this is not part of the covered material, but I'm including it in case of interest." Manuscript pages 65–71, transcribed in this section, are therefore optional reading; the covered material resumes with Chapter 5 in Section 15. □

14.1 Models and Key Properties

Source manuscript page 65

A model for systems comprised of *many, independent, unlikely* events.

One dimension — models.

- Arrivals at a Maxi self-checkout queue (timescale $\approx$ 1 minute).
- Times of radioactive decay events for 1 gram of Uranium-232 (timescale $\approx 10^{-14}$ seconds).
- Children being born in the world (timescale ≈ 0.25 seconds).

Key properties (when modelling, need to think about whether these are reasonable):

- *Independence*: an event at one time/location does not affect the probability of an event at other times/locations.
- Events don't occur simultaneously (in time or space).
- Events occur with a *rate* (often λ); e.g. in one time dimension, $\mathbb{P}\big(\text{arrival in checkout queue} \in [t, t+\delta)\big) \approx \lambda \cdot \delta$ for small δ.

14.2 Definition

Source manuscript page 66

Definition (homogeneous Poisson process)

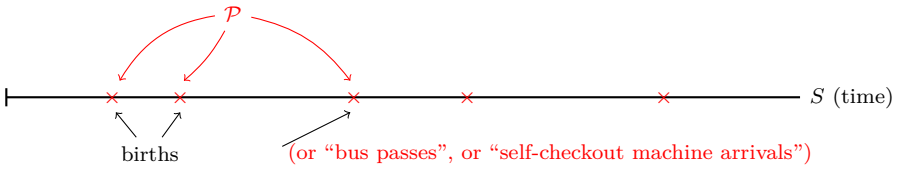
Fix an interval $S \subseteq \mathbb{R}$ (allow S unbounded, even $S = \mathbb{R}$). A *homogeneous rate-λ Poisson process in S* is a random set of points $\mathcal{P} \subseteq S$ such that, for any disjoint intervals $I_1, \ldots, I_k \subseteq S$,

(A) $|\mathcal{P} \cap I_j|$ is $\mathrm{Poi}(\lambda |I_j|)$ for $1 \leq j \leq k$;

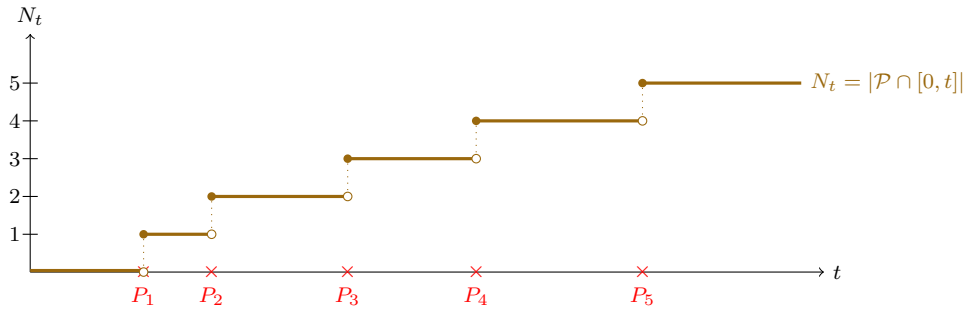
(B) $|\mathcal{P} \cap I_1|, \ldots, |\mathcal{P} \cap I_k|$ are mutually independent.

□

S usually represents *time*, and points represent *occurrences/arrivals* ("births", or "bus passes", or "self-checkout machine arrivals"):



In this case it is standard to write $N_t = |\mathcal{P} \cap [0,t]|$, the "number of arrivals by time t". $(N_t,\, t \geq 0)$ is *also* called the Poisson process:



Source manuscript page 67

Example (customers)

Customers arrive at rate 5/hr between 9 AM and 6 PM. Then

$$\mathbb{P}(\text{no customers from 9–10 AM}) = \mathbb{P}(\text{Poi}(5) = 0) = e^{-5} \approx 0.0067,$$

and

$$\begin{aligned}\mathbb{P}(&2 \text{ from 9–10},\ 3 \text{ from 10–10:30},\ 5 \text{ from 2–3:30}) \\ &= \mathbb{P}(2 \text{ from 9–10}) \cdot \mathbb{P}(3 \text{ from 10–10:30}) \cdot \mathbb{P}(5 \text{ from 2–3:30}) \\ &= \mathbb{P}(\text{Poi}(5) = 2) \cdot \mathbb{P}(\text{Poi}(2.5) = 3) \cdot \mathbb{P}(\text{Poi}(7.5) = 5) = \cdots \approx 0.0020.\end{aligned}$$

NB. Here we are modelling by a rate-5 P.P.P. on $[9, 18]$. □

Common spaces for P.P.P.: $[0,\infty)$, $\mathbb{R}$, $[a,b] \subseteq \mathbb{R}$.

14.3 From Bernoulli Trials to the Poisson Process

Source manuscript page 67 (second page numbered 19)

Note

This manuscript page is struck through with a diagonal line; its content is re-drawn as the diagram on the following page. It is transcribed for completeness. □

Poisson, Binomial, Exp, Geom, ..., Gamma. Let $X_1, X_2, X_3, \ldots$ be independent $\text{Ber}(\lambda/n)$, laid out along a line: take the first n of them, and more generally the first $m = cn$. Then

$$S_n = X_1 + \cdots + X_n \sim \text{Bin}(n, \lambda/n) \approx \text{Poi}(\lambda).$$

What about S_m, $m = tn$? Then write $\frac{\lambda}{n} = \frac{t\lambda}{tn} = \frac{t\lambda}{m}$. So

$$S_m = \text{Bin}\Big(m, \frac{\lambda}{n}\Big) = \text{Bin}\Big(m, \frac{t\lambda}{m}\Big) \approx \text{Poi}(t\lambda).$$

More generally, if $0 \le n_1 < n_2$ and $n_2 - n_1 = cn$, then

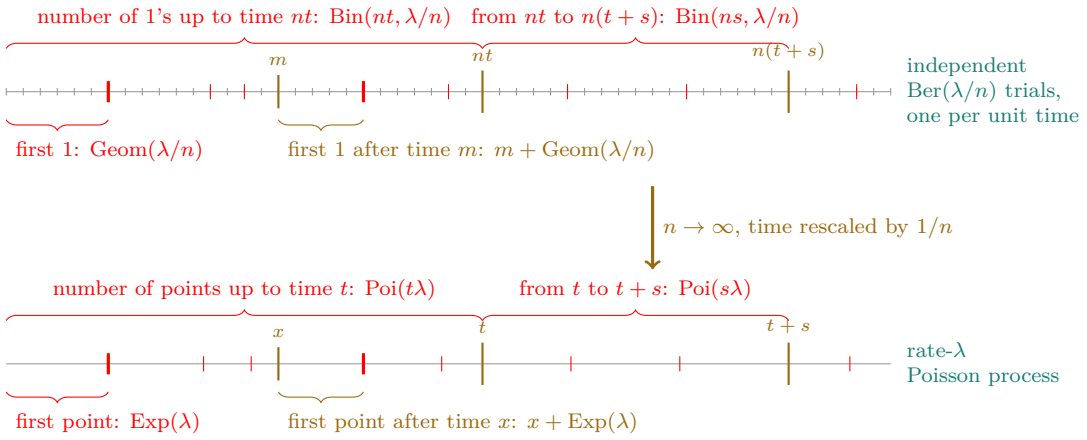
$$S_{n_2} - S_{n_1} = \text{Bin}(n_2 - n_1, p) \approx \text{Poi}(c\lambda).$$

Approximation good provided $(n_2 - n_1)p^2 \ll 1$.

Note. If $n_1 = tn$, $n_2 = (t+s)n$, then $S_{n_1} \approx \text{Poi}(t\lambda)$, $S_{n_2} - S_{n_1} \approx \text{Poi}(s\lambda)$ and $S_{n_2} \approx \text{Poi}((t+s)\lambda) \longrightarrow$ suggests $\text{Poi}(t\lambda) + \text{Poi}(s\lambda) \approx \text{Poi}((t+s)\lambda)$ for independent summands.

Source manuscript page 68

The picture: on top, independent $\text{Ber}(\lambda/n)$ trials, one per unit of time; below, the rate-λ Poisson process they become once time is rescaled by $1/n$ (so that n trials fit into one unit of time).



	Independent $\text{Ber}(\lambda/n)$ trials	Rate-λ Poisson process
number of 1's / points up to time nt / t	$\text{Bin}(nt, \lambda/n)$	$\text{Poi}(t\lambda)$
number from nt to $n(t+s)$ / from t to $t+s$	$\text{Bin}(ns, \lambda/n)$	$\text{Poi}(s\lambda)$
first 1 / first point	$\text{Geom}(\lambda/n)$	$\text{Exp}(\lambda)$
first 1 after time m / first point after time x	$m + \text{Geom}(\lambda/n)$	$x + \text{Exp}(\lambda)$

What about the time to the 2nd 1 / the 2nd point?

14.4 The Gamma Distribution

Source manuscript page 69

Let $\mathcal{P} = \{P_1, P_2, \dots\}$ be the points of a rate-λ P.P.P. on $[0, \infty)$, with $P_1 < P_2 < \cdots$. Fix $t > 0$. Then

$$\mathbb{P}(P_1 > t) = \mathbb{P}\big(|\mathcal{P} \cap [0,t]| = 0\big) = \mathbb{P}\big(\text{Poi}(\lambda t) = 0\big) = e^{-\lambda t}\frac{(\lambda t)^0}{0!} = e^{-\lambda t},$$

so $P_1 \sim \text{Exp}(\lambda)$. And

$$\mathbb{P}(P_2 > t) = \mathbb{P}\big(|\mathcal{P} \cap [0,t]| \le 1\big) = \mathbb{P}\big(\text{Poi}(\lambda t) = 0\big) + \mathbb{P}\big(\text{Poi}(\lambda t) = 1\big) = e^{-\lambda t}\frac{(\lambda t)^0}{0!} + e^{-\lambda t}\frac{(\lambda t)^1}{1!} = e^{-\lambda t}(1 + \lambda t).$$

So P_2 has CDF and PDF

$$F(t) = \begin{cases} 1 - e^{-\lambda t}(1+\lambda t), & \text{if } t \ge 0, \\ 0, & \text{if } t < 0, \end{cases} \qquad F'(t) = \begin{cases} \lambda^2 t e^{-\lambda t}, & t \ge 0, \\ 0, & t < 0, \end{cases}$$

since $\frac{d}{dt}\big(-e^{-\lambda t}(1+\lambda t)\big) = \lambda e^{-\lambda t}(1+\lambda t) - \lambda e^{-\lambda t} = \lambda e^{-\lambda t} + \lambda^2 t e^{-\lambda t} - \lambda e^{-\lambda t} = \lambda^2 t e^{-\lambda t}$.

Source manuscript page 70

For the nth point P_n, the CDF is, for $t \geq 0$,

$$F_{P_n}(t) = \mathbb{P}(P_n \leq t) = 1 - \mathbb{P}(P_n > t) = 1 - \mathbb{P}\big(|\mathcal{P} \cap [0,t]| < n\big) = 1 - \sum_{k=0}^{n-1} \mathbb{P}\big(\mathrm{Poi}(\lambda t) = k\big)$$
$$= 1 - \sum_{k=0}^{n-1} e^{-\lambda t}\frac{(\lambda t)^k}{k!} = 1 - e^{-\lambda t} - \sum_{k=1}^{n-1} e^{-\lambda t}\frac{(\lambda t)^k}{k!}.$$

The PDF is

$$f_{P_n}(t) = F'_{P_n}(t) = \lambda e^{-\lambda t} - \sum_{k=1}^{n-1}\Big(-\lambda e^{-\lambda t}\frac{(\lambda t)^k}{k!} + e^{-\lambda t}\frac{\lambda^k t^{k-1}}{(k-1)!}\Big)$$
$$= \lambda e^{-\lambda t} + \lambda e^{-\lambda t}\sum_{k=1}^{n-1}\frac{(\lambda t)^k}{k!} - \lambda e^{-\lambda t}\sum_{k=1}^{n-1}\frac{(\lambda t)^{k-1}}{(k-1)!}$$
$$= \lambda e^{-\lambda t}\frac{(\lambda t)^{n-1}}{(n-1)!} = \frac{\lambda^n t^{n-1}}{(n-1)!}e^{-\lambda t} \qquad (t \geq 0),$$

because the second sum is the first one shifted by one index, so the two sums differ by $\frac{(\lambda t)^{n-1}}{(n-1)!} - 1$, and the -1 cancels the leading $\lambda e^{-\lambda t}$.

Source manuscript page 71

Definition (Gamma function)

The *Gamma function* is

$$\Gamma : (0,\infty) \to (0,\infty), \qquad r \longmapsto \Gamma(r) = \int_0^\infty x^{r-1}e^{-x}\,dx.$$

□

Exercise. $\Gamma(n) = (n-1)!$ for $n = 1, 2, \ldots$ (integration by parts).

Definition (Gamma distribution)

A random variable X is Gamma(r,λ)-distributed if it has PDF

$$f(x) = \frac{\lambda^r x^{r-1}e^{-\lambda x}}{\Gamma(r)} \qquad \text{for } x > 0.$$

□

(So, by the exercise and the computation above, the nth point P_n of a rate-λ Poisson process is Gamma(n,λ)-distributed; in particular Gamma$(1,\lambda) = \mathrm{Exp}(\lambda)$, and $P_2 \sim$ Gamma$(2,\lambda)$ answers the question of manuscript page 68.)

Proposition

$$\int_0^\infty f(x)\,dx = 1.$$

Proof. If $\lambda = 1$ then $\displaystyle\int_0^\infty f(x)\,dx = \frac{1}{\Gamma(r)}\int_0^\infty x^{r-1}e^{-x}\,dx = 1$. If $\lambda \neq 1$, let $y = \lambda x$, $dy = \lambda\,dx$; then

$$\int_0^\infty \frac{\lambda^r x^{r-1}e^{-\lambda x}}{\Gamma(r)}\,dx = \int_0^\infty \frac{\lambda^r (y/\lambda)^{r-1}e^{-y}}{\Gamma(r)}\,\frac{dy}{\lambda} = \int_0^\infty \frac{y^{r-1}e^{-y}}{\Gamma(r)}\,dy = 1,$$

the $\lambda = 1$ case. □

15 Transforms: Moment Generating Functions

Note. The manuscript starts a new chapter here (Chapter 5), with its own page numbering; the source marks in this section refer to it.

Source manuscript: chapter 5, page 1

Definition (moment generating function)

Given a random variable X, the *moment generating function* of X is

$$M(t) = \mathbb{E}\big[e^{tX}\big].$$

□

Example

- If $X \sim \mathrm{Ber}(p)$ then $M(t) = \mathbb{E}[e^{tX}] = e^{t\cdot 1}\,\mathbb{P}(X=1) + e^{t\cdot 0}\,\mathbb{P}(X=0) = pe^t + 1 - p$.
- If $X \sim \mathrm{Exp}(\lambda)$ then

$$M(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx}\lambda e^{-\lambda x}\,dx = \int_0^\infty \lambda e^{(t-\lambda)x}\,dx = \Big[\frac{\lambda}{t-\lambda}e^{(t-\lambda)x}\Big]_0^\infty = \begin{cases}\infty, & t \geq \lambda,\\ \dfrac{\lambda}{\lambda - t}, & t < \lambda.\end{cases}$$

- If $X \sim \mathrm{Geom}(p)$ then

$$\begin{aligned} M(t) = \mathbb{E}[e^{tX}] = \sum_{k\geq 1} e^{tk}\,\mathbb{P}(X=k) &= \sum_{k\geq 1} e^{tk}(1-p)^{k-1}p = \frac{p}{1-p}\sum_{k\geq 1}\big((1-p)e^t\big)^k \\ &= \frac{p}{1-p}\cdot(1-p)e^t\sum_{k\geq 0}\big((1-p)e^t\big)^k \\ &= \begin{cases}\dfrac{pe^t}{1-(1-p)e^t}, & \text{if } (1-p)e^t < 1,\\ \infty, & \text{if } (1-p)e^t \geq 1.\end{cases}\end{aligned}$$

- If $X \sim \mathcal{N}(0,1)$ then, since $tx - \dfrac{x^2}{2} = -\dfrac{(x-t)^2}{2} + \dfrac{t^2}{2}$,

$$\mathbb{E}[e^{tX}] = \int_{\mathbb{R}} e^{tx}\varphi(x)\,dx = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{tx}e^{-x^2/2}\,dx = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-(x-t)^2/2}\,e^{t^2/2}\,dx = e^{t^2/2}.$$

□

15.1 Why the Name?

Source manuscript: chapter 5, page 2

If $M(t) = \mathbb{E}(e^{tX})$ then

$$M'(t) = \frac{d}{dt}\mathbb{E}\big(e^{tX}\big) \stackrel{?}{=} \mathbb{E}\Big(\frac{d}{dt}e^{tX}\Big) = \mathbb{E}\big[Xe^{tX}\big], \qquad M''(t) = \frac{d^2}{dt^2}\mathbb{E}\big(e^{tX}\big) \stackrel{?}{=} \mathbb{E}\Big[\frac{d^2}{dt^2}e^{tX}\Big] = \mathbb{E}\big[X^2e^{tX}\big],$$

$$\vdots$$

$$M^{(n)}(t) = \mathbb{E}\big[X^n e^{tX}\big].$$

(Here e^{tX} is the function $\Omega \times \mathbb{R} \to \mathbb{R}$, $(\omega, t) \mapsto e^{tX(\omega)}$, and "$\stackrel{?}{=}$" is the exchange of $\frac{d}{dt}$ and $\mathbb{E}$.) So

$$M'(0) = \mathbb{E}\big[X \cdot e^{0\cdot X}\big] = \mathbb{E}[X \cdot 1] = \mathbb{E}X, \qquad M''(0) = \mathbb{E}X^2, \qquad \ldots, \qquad M^{(k)}(0) = \mathbb{E}\big[X^k\big].$$

Note. If X has density f then $M(t) = \int_{\mathbb{R}} e^{tx} f(x)\,dx$; we are hoping that

$$\frac{d}{dt}\int_{\mathbb{R}} e^{tx} f(x)\,dx = \int_{\mathbb{R}} \frac{d}{dt} e^{tx} f(x)\,dx = \int_{\mathbb{R}} xe^{tx} f(x)\,dx \qquad \text{(differentiation under the integral sign).}$$

If X is discrete with finitely many possible values $x_1, \ldots, x_k$ then $M(t) = \sum_{i=1}^k e^{tx_i}\,\mathbb{P}(X = x_i)$ and $M'(t) = \sum_{i=1}^k x_i e^{tx_i}\,\mathbb{P}(X = x_i)$, so $M'(0) = \sum_{i=1}^k x_i\,\mathbb{P}(X = x_i) = \mathbb{E}X$. ✓

Proposition

If there is $r > 0$ such that $M(t) < \infty$ for $|t| < r$, then $\mathbb{E}(|X|^n) < \infty$ for all $n \in \mathbb{N}$, and for $|t| < r$

$$M(t) = \sum_{n\geq 0} \frac{\mathbb{E}\big[(tX)^n\big]}{n!} = \sum_{n\geq 0} \frac{t^n}{n!} M^{(n)}(0), \qquad M^{(k)}(t) = \sum_{n\geq k} (n)_k\, t^{n-k}\, \frac{\mathbb{E}[X^n]}{n!}.$$

In particular $M^{(k)}(0) = \mathbb{E}(X^k)$ for all $k \in \mathbb{N}$.

Can work out the moments by working out the Taylor expansion of $M(t)$.

15.2 Examples

Source manuscript: chapter 5, page 3

Example

- Ber(p): $M(t) = pe^t + (1-p)$, $M^{(k)}(t) = pe^t$, $M^{(k)}(0) = p$. So $\mathbb{E}(X^k) = p$ for all $k \geq 1$.
- Exp(λ): $M(t) = \dfrac{\lambda}{\lambda - t}$, $M'(t) = \dfrac{\lambda}{(\lambda-t)^2}$, $M''(t) = \dfrac{2\lambda}{(\lambda-t)^3}$, $\ldots$, $M^{(k)}(t) = \dfrac{k!\,\lambda}{(\lambda-t)^{k+1}}$. So $\mathbb{E}(X^k) = k!\,\lambda^{-k}$.
- Geom(p): $M(t) = \dfrac{pe^t}{1-(1-p)e^t}$. Not so nice.
- $\mathcal{N}(0,1)$:

$$M(t) = e^{t^2/2} = \sum_{n\geq 0} \frac{1}{n!}\Big(\frac{t^2}{2}\Big)^n = \sum_{n\geq 0} \frac{1}{2^n\, n!}\, t^{2n} = \sum_{\substack{k\geq 0 \\ k \text{ even}}} \frac{1}{2^{k/2}(k/2)!}\, t^k.$$

The Taylor expansion of $M(t)$ around $t=0$ is $M(t)=\sum_{k\geq 0}\frac{M^{(k)}(0)}{k!}t^k$, so, comparing coefficients,

$$\frac{M^{(k)}(0)}{k!}=\frac{1}{2^{k/2}(k/2)!}\quad\text{for } k \text{ even, i.e.}\quad \mathbb{E}\big[X^{2k}\big]=M^{(2k)}(0)=\underbrace{\frac{(2k)!}{2^k\,k!}}_{\circledast}\quad\text{for } k\geq 0,$$

and $M^{(k)}(0)=0$ for k odd. Note

$$\circledast=\frac{(2k)(2k-2)\cdots(4)(2)\cdot(2k-1)(2k-3)\cdots(3)(1)}{k(k-1)\cdots(2)(1)\cdot 2^k}=(2k-1)(2k-3)\cdots(3)(1)=(2k-1)!!\,,$$

since $(2k)(2k-2)\cdots(2)=2^k\,k!$. So

$$\mathbb{E}[X^n]=\begin{cases}(n-1)!! & \text{if } n \text{ is even},\\ 0 & \text{if } n \text{ is odd}.\end{cases}$$

□

15.3 The CDF Determines the Distribution

Source manuscript: chapter 5, page 4

Recall. The *distribution* of X is the collection of values $\{\mathbb{P}(X\in B):\ B\subseteq\mathbb{R}\}$.

Fact (key fact)

For random variables X,Y: if $F_X=F_Y$ then X and Y have the *same distribution*.

This means that the collection of values $\{\underbrace{\mathbb{P}(X\leq t)}_{F_X(t)}:\ t\in\mathbb{R}\}$ determines the distribution of X.

Notation. X and Y are *equal in distribution*, written $X\overset{d}{=}Y$, if $\mathbb{P}(X\in B)=\mathbb{P}(Y\in B)$ for all $B\subseteq\mathbb{R}$.

Example

$\Omega=\{H,T\}$ with $\mathbb{P}(\{H\})=\mathbb{P}(\{T\})=\frac{1}{2}$, and $\Omega'=\{1,\dots,6\}$ with $\mathbb{P}'(\{1\})=\mathbb{P}'(\{2\})=\cdots=\mathbb{P}'(\{6\})$. Let

$$X(\omega)=\begin{cases}1 & \omega=H\\ 0 & \omega=T\end{cases},\quad Y(\omega)=\begin{cases}1 & \omega=T\\ 0 & \omega=H\end{cases}\ (\omega\in\Omega);\qquad Z(\omega)=\begin{cases}1 & \omega \text{ even}\\ 0 & \omega \text{ odd}\end{cases}\ (\omega\in\Omega').$$

All three have the same CDF (that of $\mathrm{Ber}(\frac{1}{2})$), hence the same distribution — although $Y=1-X$ is never equal to X, and Z is defined on a different probability space altogether. □

The MGF does *not* determine the distribution of a random variable.

Example

Write $\zeta(2)=\sum_{n\geq 1}\frac{1}{n^2}$ and $\zeta(3)=\sum_{n\geq 1}\frac{1}{n^3}$. Let X have possible values $\mathbb{Z}\setminus\{0\}$, with

$$\mathbb{P}(X=k)=\frac{1}{2\zeta(2)}\cdot\frac{1}{k^2}=\mathbb{P}(X=-k)\quad(k\geq 1),\qquad\text{so}\qquad\sum_{k\geq 1}\mathbb{P}\big(X\in\{\pm k\}\big)=\frac{1}{\zeta(2)}\sum_{k\geq 1}\frac{1}{k^2}=\frac{\zeta(2)}{\zeta(2)}=1.$$

Let Y have possible values $\mathbb{Z}\setminus\{0\}$, with $\mathbb{P}(Y=k)=\mathbb{P}(Y=-k)=\dfrac{1}{2\zeta(3)}\cdot\dfrac{1}{k^3}$ $(k\geq 1)$. Then X and Y do not have the same distribution, but

$$M_X(0)=M_Y(0)=1, \qquad M_X(t)=M_Y(t)=\infty \quad \text{for } t\neq 0,$$

since for $t>0$ the terms e^{tk}/k^2 (respectively e^{tk}/k^3) with $k\to+\infty$ are unbounded, and for $t<0$ those with $k\to-\infty$ are. □

15.4 When the MGF Does Determine the Distribution; Characteristic Functions

Source manuscript: chapter 5, page 5

Fact

Let X, Y be random variables with MGFs M_X and M_Y. If there is $r>0$ such that $M_X(t)=M_Y(t)<\infty$ for $|t|<r$, then $X\overset{d}{=}Y$.

E.g.

$$X=\begin{cases}-1 & \text{w.p. } \frac{1}{3}\\ \pi & \text{w.p. } \frac{1}{6}\\ 11 & \text{w.p. } \frac{1}{2}\end{cases} \qquad\Longleftrightarrow\qquad M_X(t)=\mathbb{E}\,e^{tX}=\frac{1}{3}e^{-t}+\frac{1}{6}e^{\pi t}+\frac{1}{2}e^{11t}.$$

In particular, if X, Y are *bounded* random variables then X and Y have the same distribution $\iff M_X=M_Y$ (for a bounded random variable $M(t)<\infty$ for every t).

Definition (characteristic function)

For a random variable X, the *characteristic function* of X is

$$\varphi_X:\mathbb{R}\to\mathbb{C}, \qquad \varphi_X(t)=\mathbb{E}\big[e^{itX}\big]=\mathbb{E}\big[\cos(tX)+i\sin(tX)\big]=\mathbb{E}[\cos(tX)]+i\,\mathbb{E}[\sin(tX)].$$

□

- Disadvantage: requires $\mathbb{C}$.
- Advantage: $x\mapsto e^{itx}$ is a bounded function, so we don't have to worry about an infinite expectation.
- Advantage: uniqueness. $X\overset{d}{=}Y \iff \varphi_X\equiv\varphi_Y$.

16 Functions of a Random Variable

Source manuscript: chapter 5, page 6

If $X=\text{Unif}\{-1,0,1,2\}$ and $Y=X^2$, what is the distribution of Y? Possible values for Y: $(-1)^2=1$, $0^2=0$, $1^2=1$, $2^2=4$, and

$$\begin{aligned}&\mathbb{P}(Y=0)=\mathbb{P}(X^2=0)=\mathbb{P}(X=0)=\tfrac{1}{4},\\ &\mathbb{P}(Y=1)=\mathbb{P}(X=1 \text{ or } X=-1)=\tfrac{1}{2},\\ &\mathbb{P}(Y=4)=\mathbb{P}(X^2=4)=\mathbb{P}(X=2 \text{ or } X=-2)=\mathbb{P}(X=2)=\tfrac{1}{4}.\end{aligned}$$

Proposition (functions of a discrete random variable)

For a *discrete* random variable X with PMF p_X and possible values S_X, if $Y = g(X)$ then Y has possible values

$$S_Y = \{y : \exists x \in S_X \text{ s.t. } g(x) = y\} = \{g(x) : x \in S_X\}$$

and PMF p_Y given by

$$p_Y(y) = \mathbb{P}\big(g(X) = y\big) = \mathbb{P}\Big(\bigcup_{\substack{x \in S_X \\ g(x)=y}} \{X = x\}\Big) = \sum_{\substack{x \in S_X \\ g(x)=y}} \mathbb{P}(X = x) = \sum_{\substack{x \in S_X \\ g(x)=y}} p_X(x) \qquad \text{for } y \in S_Y.$$

Example

$X = \text{Unif}\{-1, 0, 1, 2\}$, $Y = 2X^3$. Then $S_Y = \{-2, 0, 2, 16\}$ and $p_Y(-2) = p_Y(0) = p_Y(2) = p_Y(16) = \frac{1}{4}$. □

16.1 Functions of a Continuous Random Variable

Source manuscript: chapter 5, page 7

For a *continuous* random variable X there are more possibilities. With $X = \text{Unif}[-1, 1]$:

$$\begin{array}{lll} Y = X^2 & \longrightarrow \text{continuous} & g(x) = x^2 \\ Y = \mathbf{1}_{X>0} & \longrightarrow \text{discrete} & g(x) = \mathbf{1}_{x>0} \\ Y = \max(X, 0) & \longrightarrow \text{neither} & g(x) = \max(x, 0) \end{array}$$

But we can still find the distribution of Y based on that of X.

Example

$U \sim \text{Unif}[0, 1]$, $g(x) = -\ln(1 - x)$, $Y = g(U)$. Then for $t \geq 0$,

$$\mathbb{P}(Y \leq t) = \mathbb{P}\big(-\ln(1 - U) \leq t\big) = \mathbb{P}\big(\ln(1 - U) \geq -t\big) = \mathbb{P}\big(1 - U \geq e^{-t}\big) = \mathbb{P}\big(U \leq 1 - e^{-t}\big) = 1 - e^{-t}.$$

So Y is Exp(1)-distributed. □

Exercise. If $g(x) = -\frac{1}{\lambda}\ln(1 - x)$ then $g(U)$ is Exp(λ)-distributed.

Source manuscript: chapter 5, page 8

Example

$Z \sim \mathcal{N}(0, 1)$, $g(x) = x^2$, $Y = g(Z)$. For $t \geq 0$,

$$\mathbb{P}(Y \leq t) = \mathbb{P}(Z^2 \leq t) = \mathbb{P}\big(-\sqrt{t} \leq Z \leq \sqrt{t}\big) = \Phi(\sqrt{t}) - \Phi(-\sqrt{t}) = 2\,\Phi(\sqrt{t}) - 1.$$

So, for $t > 0$,

$$f_Y(t) = \frac{d}{dt}\mathbb{P}(Y \leq t) = \frac{d}{dt}\big(2\,\Phi(\sqrt{t}) - 1\big) = \frac{2\,\Phi'(\sqrt{t})}{2\sqrt{t}} = \frac{1}{\sqrt{t}} \cdot \frac{1}{\sqrt{2\pi}} e^{-(\sqrt{t})^2/2} = \frac{1}{\sqrt{2\pi t}}\, e^{-t/2}.$$

□

Definition (Gamma distribution)

A random variable G is Gamma(r,λ)-distributed if it has density

$$f_G(x) = \begin{cases} \dfrac{1}{\Gamma(r)}\,\lambda^r x^{r-1} e^{-\lambda x} & \text{if } x>0,\\ 0 & \text{otherwise,}\end{cases} \qquad \text{where } \Gamma(r)=\int_0^\infty x^{r-1}e^{-x}\,dx.$$

□

With $r=\lambda=\frac12$ this is

$$f_G(x) = \frac{1}{\Gamma(\frac12)}\cdot\frac{1}{\sqrt{2x}}\,e^{-x/2} = \frac{\sqrt{\pi}}{\Gamma(\frac12)}\,f_Y(x) \qquad (x>0).$$

Then

$$1=\int_0^\infty f_G(x)\,dx = \frac{\sqrt{\pi}}{\Gamma(\frac12)}\int_0^\infty f_Y(x)\,dx = \frac{\sqrt{\pi}}{\Gamma(\frac12)}, \qquad \text{so} \qquad \Gamma(\tfrac12)=\sqrt{\pi} \quad\text{and}\quad f_Y=f_G,$$

so $Y=Z^2$ is Gamma$(\frac12,\frac12)$-distributed.

16.2 Affine Transformations

Source manuscript: chapter 5, page 9

Suppose X is continuous with density f_X and $Y=aX+b$ with $a>0$. Then

$$F_Y(y)=\mathbb{P}(Y\le y)=\mathbb{P}(aX+b\le y)=\mathbb{P}\big(X\le (y-b)/a\big)=F_X\big((y-b)/a\big),$$

so

$$f_Y(y)=F_Y'(y)=\frac{d}{dy}F_X\big((y-b)/a\big)=\frac{1}{a}\,f_X\big((y-b)/a\big).$$

For $a<0$: $F_Y(y)=\mathbb{P}(aX+b\le y)=\mathbb{P}\big(X\ge (y-b)/a\big)=1-F_X\big((y-b)/a\big)$, so

$$f_Y(y)=-\frac{1}{a}\,f_X\Big(\frac{y-b}{a}\Big)=\frac{1}{|a|}\,f_X\Big(\frac{y-b}{a}\Big).$$

Proposition

If X has density f_X and $Y=aX+b$ with $a\neq 0$, then

$$f_Y(y)=\frac{1}{|a|}\,f_X\Big(\frac{y-b}{a}\Big).$$

Example

$X\sim\mathcal{N}(\mu,\sigma^2)$, $f_X(x)=\dfrac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/2\sigma^2}$. For $Y=aX+b$,

$$f_Y(y)=\frac{1}{|a|}\,f_X\Big(\frac{y-b}{a}\Big)=\frac{1}{\sqrt{2\pi a^2\sigma^2}}\exp\Big(-\Big(\frac{y-b}{a}-\mu\Big)^2\Big/2\sigma^2\Big)=\frac{1}{\sqrt{2\pi(a\sigma)^2}}\exp\Big(-\frac{(y-a\mu-b)^2}{2(a\sigma)^2}\Big),$$

so $Y\sim\mathcal{N}\big(a\mu+b,(a\sigma)^2\big)$. □

16.3 The "General" Formula for $Y = g(X)$

Source manuscript: chapter 5, page 10

Setting. $g : \mathbb{R} \to \mathbb{R}$ differentiable and injective, with $g'(x) = 0$ for at most finitely many x; X continuous, $Y = g(X)$. If g is differentiable and injective then g is monotone.

Case 1: $g \uparrow$. Then

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\big(g(X) \le y\big) = \mathbb{P}\big(X \le g^{-1}(y)\big) = F_X\big(g^{-1}(y)\big),$$

so

$$f_Y(y) = F_Y'(y) = \frac{d}{dy} F_X\big(g^{-1}(y)\big) = \underbrace{F_X'\big(g^{-1}(y)\big)}_{f_X(g^{-1}(y))} \cdot \underbrace{\frac{d}{dy} g^{-1}(y)}_{1/g'(g^{-1}(y))} = f_X\big(g^{-1}(y)\big) \cdot \frac{1}{g'\big(g^{-1}(y)\big)},$$

where the second factor comes from $g(g^{-1}(y)) = y$: differentiating with the chain rule,

$$1 = \frac{d}{dy}\, g\big(g^{-1}(y)\big) = g'\big(g^{-1}(y)\big) \cdot \frac{d}{dy} g^{-1}(y) \qquad \text{if } g'(g^{-1}(y)) \neq 0;\ \text{otherwise } g^{-1} \text{ is not differentiable at } y!$$

Case 2: $g \downarrow$. Then a similar computation gives

$$f_Y(y) = -f_X\big(g^{-1}(y)\big) \cdot \frac{1}{g'\big(g^{-1}(y)\big)}.$$

Fact (conclusion; Fact 5.24)

Let X be a continuous random variable with density f_X. If $g : \mathbb{R} \to \mathbb{R}$ is differentiable and injective, with $g'(x) = 0$ for at most finitely many x, and $Y = g(X)$, then

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\, \frac{1}{\big|g'\big(g^{-1}(y)\big)\big|} \qquad \text{for } y \text{ such that } g^{-1}(y) \text{ exists and } g'\big(g^{-1}(y)\big) \neq 0$$

(for the other points, set $f_Y(y) = 0$).

Source manuscript: chapter 5, page 11

Fact (Fact 5.27)

If X has density f_X, g is differentiable with $\big|\{x :\ g'(x) = 0\}\big| < \infty$, and $Y = g(X)$, then

$$f_Y(y) = \sum_{\substack{x:\ g(x)=y \\ g'(x)\neq 0}} f_X(x) \cdot \frac{1}{|g'(x)|}, \qquad \text{and} \qquad f_Y(y) = 0 \ \text{ if there is no } x \text{ with } g(x) = y \text{ and } g'(x) \neq 0.$$

Example

$X \sim \mathrm{Exp}(1)$, $Y = (X-1)^2$: $g(x) = (x-1)^2$, $g'(x) = 2(x-1)$, and $g(x) = y \iff x = 1 + \sqrt{y}$ or $x = 1 - \sqrt{y}$ (the same if $y = 0$). So for $y > 0$,

$$f_Y(y) = \frac{f_X(1+\sqrt{y})}{|g'(1+\sqrt{y})|} + \underbrace{\frac{f_X(1-\sqrt{y})}{|g'(1-\sqrt{y})|}}_{0 \text{ if } y>1} = \begin{cases} \dfrac{e^{-(1+\sqrt{y})}}{2\sqrt{y}} + \dfrac{e^{-(1-\sqrt{y})}}{2\sqrt{y}} & \text{if } 0 < y \le 1, \\ \dfrac{e^{-(1+\sqrt{y})}}{2\sqrt{y}} & \text{if } y > 1, \end{cases}$$

since $f_X(x) = e^{-x}\mathbf{1}_{x\ge 0}$ vanishes at $1 - \sqrt{y} < 0$ when $y > 1$, and $|g'(1 \pm \sqrt{y})| = 2\sqrt{y}$. □

17 Joint Distributions

Source manuscript: chapter 6, page 1

Joint distributions are just the distributions of random *vectors* ≡ multiple random variables defined on the same space. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $\vec{X} : \Omega \to \mathbb{R}^2$, $\vec{X}(\omega) = \big(X_1(\omega), X_2(\omega)\big)$, then to specify the distribution of $\vec{X}$ we must know $\mathbb{P}\big((X_1, X_2) \in B\big)$ for $B \subseteq \mathbb{R}^2$.

Example (three coin flips)

Flip a coin 3 times; let $X = \mathbf{1}_{\{\text{first flip tails}\}}$, so $S_X = \{0, 1\}$, and Y = total number of tails, so $S_Y = \{0, 1, 2, 3\}$. Then

$$S_{X,Y} = \{(0,0), (0,1), (0,2), (1,1), (1,2), (1,3)\} \qquad \text{with probabilities} \qquad \tfrac{1}{8},\ \tfrac{1}{4},\ \tfrac{1}{8},\ \tfrac{1}{8},\ \tfrac{1}{4},\ \tfrac{1}{8},$$

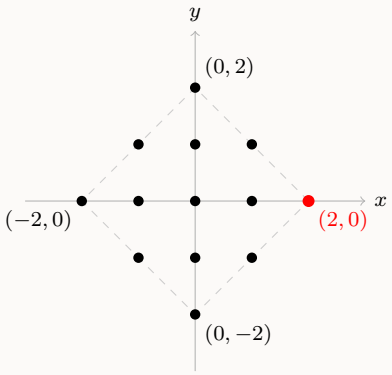
as the table of outcomes shows:

	$Y=0$	$Y=1$	$Y=2$	$Y=3$
$X=0$	HHH	$HHT,\ HTH$	HTT	
$X=1$		THH	$THT,\ TTH$	TTT

□

Example

Pick a grid point (X, Y) uniformly at random from the diamond in the figure. If $X = 2$ then $Y = 0$.



□

Definition (joint PMF (Def. 1))

If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $X_1, \ldots, X_n : \Omega \to \mathbb{R}$ are *discrete* random variables, then the *joint PMF* of $X_1, \ldots, X_n$ is

$$p(x_1, \ldots, x_n) = \mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) \qquad \text{for } x_1 \in S_{X_1}, \ldots, x_n \in S_{X_n}.$$

(NB: can have $p(x_1, \ldots, x_n) = 0$ for some points — see the examples above.) □

Exercise. $\sum p(x_1, \ldots, x_n) = 1$, where the sum is over $(x_1, \ldots, x_n) \in S_{X_1} \times \cdots \times S_{X_n}$.

Definition (Def. 2)

For $X_1, \dots, X_n$ as in Def. 1, if $g : \mathbb{R}^n \to \mathbb{R}$ and

$$\sum_{(x_1,\dots,x_n)\in S_{X_1}\times\cdots\times S_{X_n}} |g(x_1,\dots,x_n)|\, p(x_1,\dots,x_n) < \infty,$$

then define

$$\mathbb{E}\big[g(X_1,\dots,X_n)\big] = \sum_{(x_1,\dots,x_n)\in S_{X_1}\times\cdots\times S_{X_n}} g(x_1,\dots,x_n)\, p(x_1,\dots,x_n).$$

□

Source manuscript: chapter 6, page 2

Definition (marginal PMFs (Def. 3))

For $X_1, \dots, X_n$ as in Def. 1, the PMFs of $X_1, \dots, X_n$ are called the *marginal* PMFs. □

Example

For the three-coin-flip example ($X = \mathbf{1}_{\{\text{first flip tails}\}}$, $Y =$ total number of tails), the marginal PMF of X is

$$p_X(0) = \mathbb{P}(X=0) = \mathbb{P}\big((X,Y)=(0,0)\big) + \mathbb{P}\big((X,Y)=(0,1)\big) + \mathbb{P}\big((X,Y)=(0,2)\big) = \tfrac{1}{8}+\tfrac{1}{4}+\tfrac{1}{8} = \tfrac{1}{2},$$
$$p_X(1) = \cdots = \tfrac{1}{2}.$$

□

Fact

If $X_1, \dots, X_n$ are discrete with joint PMF $p = p_{X_1,\dots,X_n}$, then for $1 \le j \le n$ the marginal PMF of X_j is

$$p_{X_j}(x) = \sum_{\substack{x_1\in S_{X_1},\ \dots,\ x_{j-1}\in S_{X_{j-1}} \\ x_{j+1}\in S_{X_{j+1}},\ \dots,\ x_n\in S_{X_n}}} p(x_1,\dots,x_{j-1},x,x_{j+1},\dots,x_n).$$

Proof. The event $\{X_j = x\}$ is the disjoint union, over all $(x_1,\dots,x_{j-1},x_{j+1},\dots,x_n)$, of the events $\{X_1 = x_1,\dots,X_{j-1}=x_{j-1},\ X_j = x,\ X_{j+1}=x_{j+1},\dots,X_n=x_n\}$, so

$$p_{X_j}(x) = \mathbb{P}(X_j = x) = \sum_{x_1,\dots,x_{j-1},x_{j+1},\dots,x_n} \mathbb{P}(X_1=x_1,\dots,X_{j-1}=x_{j-1},X_j=x,X_{j+1}=x_{j+1},\dots,X_n=x_n)$$
$$= \text{RHS}.$$

□

For $n = 2$, (X,Y), this reduces to $p_X(x) = \sum_{y\in S_Y} p(x,y)$ and $p_Y(y) = \sum_{x\in S_X} p(x,y)$.

Example

X_1, X_2 independent Unif$\{1,2,3,4\}$; $Y_1 = \min(X_1,X_2)$, $Y_2 = |X_1 - X_2|$.

a) Joint PMF of X_1, X_2: $p_{X_1,X_2}(x_1,x_2) = \frac{1}{16}$ for $x_1 \in [4]$, $x_2 \in [4]$.

b) Possible values of (Y_1,Y_2): $(1,0),(1,1),(1,2),(1,3)$; $(2,0),(2,1),(2,2)$; $(3,0),(3,1)$; $(4,0)$.

Source manuscript: chapter 6, page 3

c) $p_{Y_1,Y_2}(y_1,y_2) = \mathbb{P}\big(\min(X_1,X_2)=y_1,\ \max(X_1,X_2)=y_1+y_2\big)$. If $y_2 = 0$ this is $\mathbb{P}(X_1 = X_2 =$

$y_1) = \frac{1}{16}$. If $y_2 > 0$ then this is $\mathbb{P}\big((X_1, X_2) = (y_1, y_1 + y_2)$ or $(X_1, X_2) = (y_1 + y_2, y_1)\big) = \frac{1}{8}$.

d) p_{Y_1}? $p_{Y_1}(1) = \frac{1}{16} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{7}{16}$, $p_{Y_1}(2) = \frac{5}{16}$, $p_{Y_1}(3) = \frac{3}{16}$, $p_{Y_1}(4) = \frac{1}{16}$;
$p_{Y_2}(0) = \frac{1}{4}$, $p_{Y_2}(1) = \frac{3}{8}$, $p_{Y_2}(2) = \frac{1}{4}$, $p_{Y_2}(3) = \frac{1}{8}$.

e) $\mathbb{P}(Y_1 Y_2 \le 2) = \mathbb{P}\big((Y_1, Y_2) \in \{(1,0), (1,1), (1,2), (2,0), (2,1), (3,0), (4,0)\}\big) = \frac{10}{16}$.

f)

$$\begin{aligned}\mathbb{E}[Y_1 Y_2] &= \sum_{y_1=1}^{4} \sum_{y_2=0}^{3} y_1 y_2 \, p_{Y_1,Y_2}(y_1, y_2) \\ &= 1 \cdot p(1,1) + 2\big(p(1,2) + p(2,1)\big) + 3\big(p(3,1) + p(1,3)\big) + 4\, p(2,2) \\ &= \frac{1}{8} + 2\Big(\frac{1}{8} + \frac{1}{8}\Big) + 3\Big(\frac{1}{8} + \frac{1}{8}\Big) + 4 \cdot \frac{1}{8} = \frac{15}{8}.\end{aligned}$$

□

17.1 The Multinomial Distribution

New distribution: the multinomial. Roll a die n times. Let N_1 = number of times 1 is rolled, …, N_6 = number of times 6 is rolled. Each of $N_1, \dots, N_6$ is $\text{Bin}(n, \frac{1}{6})$, but they are not independent.

$$\mathbb{P}\big((N_1, \dots, N_6) = (n_1, \dots, n_6)\big) = \binom{n}{n_1, \dots, n_6} \frac{1}{6^n} \quad \text{if } n_1 + \cdots + n_6 = n, \qquad = 0 \quad \text{otherwise.}$$

Source manuscript: chapter 6, page 4

Definition (multinomial distribution)

For $1 \le r \le n$ and $p_1, \dots, p_r \ge 0$ with $p_1 + \cdots + p_r = 1$, a random vector $\vec{X} = (X_1, \dots, X_r)$ is $\text{Mult}(n, r, p_1, \dots, p_r)$-distributed if

$$S_{\vec{X}} = \{(n_1, \dots, n_r) \in \mathbb{Z}^r_{\ge 0} : \; n_1 + \cdots + n_r = n\}$$

and

$$\mathbb{P}\big(\vec{X} = (n_1, \dots, n_r)\big) = \binom{n}{n_1, \dots, n_r} p_1^{n_1} \cdots p_r^{n_r} \qquad \text{for } (n_1, \dots, n_r) \in S_{\vec{X}}.$$

□

NB. When $r = 2$ this means $\mathbb{P}\big((X_1, X_2) = (k, n-k)\big) = \binom{n}{k, n-k} p_1^k p_2^{n-k} = \binom{n}{k} p^k (1-p)^{n-k}$: the binomial.

Example

Urn with 1 green, 2 red, 3 yellow. Sample 10 times (with replacement). Then

$$\mathbb{P}(3\,\text{g},\ 2\,\text{r},\ 5\,\text{y}) = \binom{10}{3, 2, 5} \Big(\frac{1}{6}\Big)^3 \Big(\frac{1}{3}\Big)^2 \Big(\frac{1}{2}\Big)^5.$$

□

Example

Roll a die 100 times.

$$\mathbb{P}(22 \text{ ones},\ 17 \text{ fives}) = \sum_{\substack{n_1,\dots,n_6:\ n_1+\cdots+n_6=100 \\ n_1=22,\ n_5=17}} \binom{100}{n_1,\dots,n_6} \left(\frac{1}{6}\right)^{100}.$$

Another way to compute it: view the outcomes $\{2,3,4,6\}$ as "other". Then

$$\mathbb{P}(22 \text{ ones},\ 17 \text{ fives},\ 61 \text{ other}) = \binom{100}{22,17,61} \left(\frac{1}{6}\right)^{22} \left(\frac{1}{6}\right)^{17} \left(\frac{4}{6}\right)^{61}.$$

□

17.2 Joint Densities

Source manuscript: chapter 6, page 5

Definition (joint density)

If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $X_1, \dots, X_n : \Omega \to \mathbb{R}$ are *continuous* random variables, then we say $f : \mathbb{R}^n \to [0, \infty)$ is a *joint density function* of $X_1, \dots, X_n$ if f is such that, for $B \subseteq \mathbb{R}^n$,

$$\mathbb{P}\big((X_1, \dots, X_n) \in B\big) = \int \cdots \int_B f(x_1, \dots, x_n)\, dx_1 \cdots dx_n.$$

Write $f = f_{X_1,\dots,X_n}$. □

If f is a joint density of $X_1, \dots, X_n$ and $g : \mathbb{R}^n \to \mathbb{R}$ is such that

$$\int \cdots \int_{\mathbb{R}^n} f(x_1, \dots, x_n)\, |g(x_1, \dots, x_n)|\, dx_1 \cdots dx_n < \infty,$$

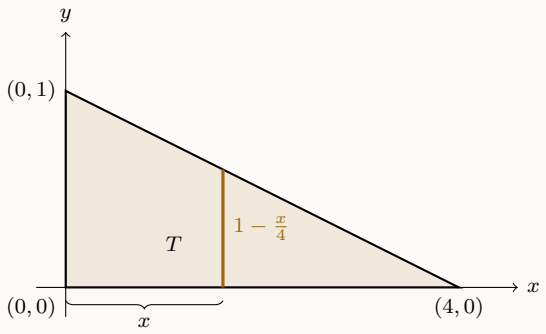
then

$$\mathbb{E}\, g(X_1, \dots, X_n) = \int \cdots \int_{\mathbb{R}^n} f(x_1, \dots, x_n)\, g(x_1, \dots, x_n)\, dx_1 \cdots dx_n.$$

Example (uniform point in a triangle)

$(X, Y) \to$ uniform point in the triangle T with corners $(0,0)$, $(4,0)$, $(0,1)$. Density:

$$f(x,y) = \frac{1}{\text{Area}(T)} \cdot \mathbf{1}_{(x,y)\in T} = \frac{1}{2} \cdot \mathbf{1}_{(x,y)\in T}.$$



If $g(x,y) = x$ then, since the vertical section of T at $x \in [0,4]$ is $0 \le y \le 1 - \frac{x}{4}$,

$$\mathbb{E}\, g(X,Y) = \iint_{\mathbb{R}^2} x \cdot \tfrac{1}{2}\mathbf{1}_{(x,y)\in T}\, dy\, dx = \int_0^4 \int_0^{1-x/4} \frac{x}{2}\, dy\, dx = \int_0^4 \Big(1 - \frac{x}{4}\Big)\frac{x}{2}\, dx$$
$$= \left[\int_0^4 \frac{x}{2}\, dx - \int_0^4 \frac{x^2}{8}\, dx\right] = \frac{16}{4} - \frac{64}{3\cdot 8} = \frac{4}{3}.$$

Source manuscript: chapter 6, page 6

What is $\mathbb{P}(X \le Y)$? Use $\mathbb{P}(E) = \mathbb{E}[\mathbf{1}_E] = 1 \cdot \mathbb{P}(\mathbf{1}_E = 1) = 1 \cdot \mathbb{P}(E)$. So take $g(x,y) = \mathbf{1}_{x\le y}$. Then

$$\mathbb{P}(X \le Y) = \mathbb{E}\,\mathbf{1}_{X\le Y} = \iint_{\{(x,y)\in\mathbb{R}^2:\ x\le y\}} f(x,y)\, dy\, dx = \iint_{x\le y} \tfrac{1}{2}\mathbf{1}_{(x,y)\in T}\, dy\, dx$$
$$= \frac{1}{2}\int_0^4 \int_{\mathbb{R}} \mathbf{1}_{x\le y\le 1-x/4}\, dy\, dx = \frac{1}{2}\int_0^{4/5} \Big(1 - \frac{5x}{4}\Big) dx$$
$$= \frac{1}{2}\Big[x - \frac{5x^2}{8}\Big]_0^{4/5} = \frac{1}{2}\Big[\frac{4}{5} - \frac{5}{8}\cdot\frac{16}{25}\Big] = \frac{1}{2}\Big[\frac{4}{5} - \frac{2}{5}\Big] = \frac{1}{5},$$

since $x \le 1 - \frac{x}{4}$ exactly when $x \le \frac{4}{5}$.
Let A be the area of the rectangle with corners $(0,0)$, (X,Y), $(X,0)$, $(0,Y)$. Then

$$\mathbb{E}A = \mathbb{E}[XY] = \iint_T xy\, f(x,y)\, dy\, dx = \int_0^4 \int_0^{1-x/4} xy \cdot \frac{1}{2}\, dy\, dx = \int_0^4 \frac{x(4-x)^2}{64}\, dx$$
$$\overset{z=x/4}{=} \frac{16}{2}\int_0^1 \int_0^{1-z} zy\, dy\, dz = \frac{16}{2}\int_0^1 \frac{z(1-z)^2}{2}\, dz$$
$$= 4\left[\int_0^1 z - 2z^2 + z^3\, dz\right] = 4\Big[\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\Big] = \frac{1}{3}.$$

□

Source manuscript: chapter 6, page 7

Proposition (marginal densities)

If $X_1, \dots, X_n$ have joint density f then for each $1 \le j \le n$, X_j has density

$$f_{X_j}(x) = \underbrace{\int_{\mathbb{R}} \cdots \int_{\mathbb{R}}}_{n-1 \text{ times}} f(x_1, \dots, x_{j-1}, x, x_{j+1}, \dots, x_n)\, dx_1 \cdots dx_{j-1}\, dx_{j+1} \cdots dx_n.$$

$\big[\, n = 2,\ (X,Y) \ \to\ f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\, dy \,\big]$

Proof ($n = 2$ only).

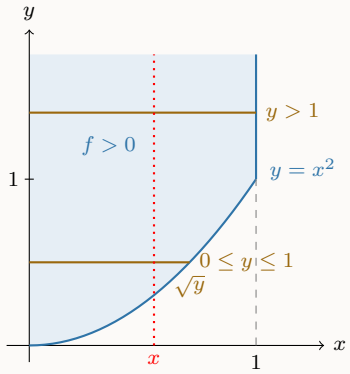
$$\mathbb{P}(X \le b) = \mathbb{P}(X \le b,\ Y \in \mathbb{R}) = \iint \mathbf{1}_{(-\infty,b]\times\mathbb{R}}(x,y)\, f(x,y)\, dy\, dx = \int_{-\infty}^b \int_{\mathbb{R}} f(x,y)\, dy\, dx.$$

So $\frac{d}{db}\mathbb{P}(X \le b) = \frac{d}{db}\int_{-\infty}^b \int_{\mathbb{R}} f(x,y)\, dy\, dx = \int_{\mathbb{R}} f(b,y)\, dy$, so $\int_{\mathbb{R}} f(x,y)\, dy$ is a density for X. □

Example

(X, Y) with joint density

$$f(x,y) = \begin{cases} 2xe^{x^2-y} & \text{if } 0 < x < 1,\ y > x^2, \\ 0 & \text{otherwise.} \end{cases}$$



Then $\int_0^1 \int_{x^2}^{\infty} 2xe^{x^2-y}\,dy\,dx = \int_0^1 2x\,dx = 1$, so f is a joint density.

$$f_X(x) = \int_{x^2}^{\infty} 2xe^{x^2-y}\,dy = 2x \qquad \text{for } 0 < x < 1.$$

$f_Y(y)$? If $y > 1$ then $y > x^2$ for all $x \in [0,1]$, so

$$\int_0^1 f(x,y)\,dx = \int_0^1 2xe^{x^2-y}\,dx = e^{-y}\left[e^{x^2}\right]_0^1 = e^{-y}(e-1).$$

If $0 \le y \le 1$ then we need $0 < x \le \sqrt{y}$, so

$$\int_{\mathbb{R}} f(x,y)\,dx = \int_0^{\sqrt{y}} 2xe^{x^2-y}\,dx = e^{-y}\left[e^{x^2}\right]_0^{\sqrt{y}} = e^{-y}\left(e^y - 1\right) = 1 - e^{-y}.$$

So

$$f_Y(y) = \begin{cases} e^{-y}(e-1) & \text{if } y > 1, \\ 1 - e^{-y} & \text{if } 0 \le y \le 1, \\ 0 & \text{otherwise} \end{cases}$$

is a density for Y. We have e.g.

$$\begin{aligned} \mathbb{P}(Y < 3X^2) = \mathbb{P}(X^2 \le Y < 3X^2) &= \int_0^1 \int_{x^2}^{3x^2} 2xe^{x^2-y}\,dy\,dx = \int_0^1 2x\left(1 - e^{-2x^2}\right)dx \\ &= 1 - \left[-\tfrac{1}{2}e^{-2x^2}\right]_0^1 = \frac{1}{2}\left(1 + \frac{1}{e^2}\right). \end{aligned}$$

□

Source manuscript: chapter 6, page 8

Note

Some pairs (X, Y) of continuous random variables have no joint distribution — that is, no joint density. □

Example

If X has density f_X and $Y = X$, then for *any* non-negative function $f : \mathbb{R}^2 \to [0, \infty)$,

$$\iint_{\{y=x\}} f(x,y)\,dy\,dx = \int_{-\infty}^{\infty}\int_{y}^{y} f(x,y)\,dy\,dx = \int_{-\infty}^{\infty} 0\,dx = 0 \;\neq\; \mathbb{P}(Y = X) = 1,$$

so f is not a joint density function for (X, Y). □

The same calculation shows that if a pair (X, Y) *does* have a joint density f, then $\mathbb{P}(X = Y) = \mathbb{P}\big((X,Y) \in L\big) = 0$, where $L = \{(x,y) \in \mathbb{R}^2 : \; y = x\}$. More generally, if (X, Y) have a joint density and $A \subseteq \mathbb{R}^2$ is any set with area 0, then $\mathbb{P}\big((X,Y) \in A\big) = 0$.

17.3 Joint Distributions and Independence

Definition (joint CDF)

The *joint CDF* of $(X_1, \ldots, X_n)$ is

$$F(x_1, \ldots, x_n) = \mathbb{P}(X_1 \le x_1, \ldots, X_n \le x_n).$$

□

Fact

- Discrete random variables $X_1, \ldots, X_n$ are independent $\iff$ their joint PMF is $p(x_1, \ldots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n)$.
- If continuous random variables $X_1, \ldots, X_n$ (defined on a common probability space) are independent, then they have a joint PDF $f(x_1, \ldots, x_n)$, and $f(x_1, \ldots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$.
- If continuous random variables $X_1, \ldots, X_n$ have a joint PDF $f(x_1, \ldots, x_n)$ and $f(x_1, \ldots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$, then $X_1, \ldots, X_n$ are independent.

⊛ $X_1, \ldots, X_n$ are independent $\iff$ $F(x_1, \ldots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n)$ for all $x_1, \ldots, x_n \in \mathbb{R}$ $\iff$ $\mathbb{P}(X_1 > x_1, \ldots, X_n > x_n) = \mathbb{P}(X_1 > x_1) \cdots \mathbb{P}(X_n > x_n)$ for all $x_1, \ldots, x_n \in \mathbb{R}$.

Exercise. Verify ⊛ when $X_1, \ldots, X_n$ are independent and discrete.

17.4 Uniform Distributions on Regions

Source manuscript: chapter 6, page 9

For a region $R \subseteq \mathbb{R}^d$, denote by $\mathrm{Vol}(R)$ the d-dimensional volume of R: for $d = 2$, $\mathrm{Vol}(R)$ is the area of R; for $d = 1$, $\mathrm{Vol}(R)$ is the length of R.

Definition (uniform distribution on a region)

Say $\vec{X} = (X_1, \ldots, X_d)$ is *uniformly distributed* over $R \subseteq \mathbb{R}^d$ (a region with $0 < \mathrm{Vol}(R) < \infty$), and write $\vec{X} \sim \mathrm{Unif}(R)$, if

$$\mathbb{P}(\vec{X} \in B) = \frac{\mathrm{Vol}(B)}{\mathrm{Vol}(R)} \qquad \text{for } B \subseteq R.$$

□

If $\vec{X} \sim \text{Unif}(R)$ then $X_1, \ldots, X_d$ have joint density

$$f(x_1, \ldots, x_d) = \begin{cases} \dfrac{1}{\text{Vol}(R)} & \text{if } (x_1, \ldots, x_d) \in R, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{i.e.} \qquad f(\vec{x}) = \frac{1}{\text{Vol}(R)} \mathbf{1}_R(\vec{x}),$$

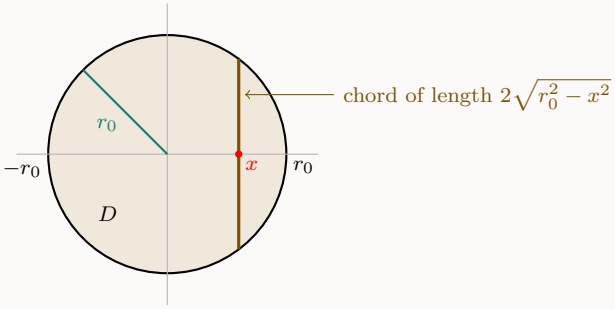
since, then, for $B \subseteq R$,

$$\int \cdots \int_B f(x_1, \ldots, x_d)\, dx_1 \cdots dx_d = \int \cdots \int_B \frac{1}{\text{Vol}(R)}\, dx_1 \cdots dx_d = \frac{\text{Vol}(B)}{\text{Vol}(R)} = \mathbb{P}(\vec{X} \in B)$$

(for a general $B \subseteq \mathbb{R}^d$, apply this to $B \cap R$, as $\mathbb{P}(\vec{X} \notin R) = 0$).

Example (darts)

$(X, Y) \sim \text{Unif}(D)$, where $D = B(\vec{0}, r_0) = \{(x, y) \in \mathbb{R}^2 : \sqrt{x^2 + y^2} \le r_0\}$.



Joint density: $f(x, y) = \mathbf{1}_{x^2+y^2 \le r_0^2} \cdot \dfrac{1}{\text{Vol}(D)} = \mathbf{1}_{x^2+y^2 \le r_0^2} \cdot \dfrac{1}{\pi r_0^2}$.

Marginal densities: for $|x| \le r_0$ the vertical section of D at x is the chord $|y| \le \sqrt{r_0^2 - x^2}$, so

$$f_X(x) = \int_{\mathbb{R}} \mathbf{1}_{x^2+y^2 \le r_0^2} \cdot \frac{1}{\pi r_0^2}\, dy = \begin{cases} \dfrac{2\sqrt{r_0^2 - x^2}}{\pi r_0^2} & \text{if } |x| \le r_0, \\ 0 & \text{otherwise,} \end{cases}$$

and likewise

$$f_Y(y) = \frac{2\sqrt{r_0^2 - y^2}}{\pi r_0^2} \mathbf{1}_{[-r_0, r_0]}(y).$$

□

17.5 Verifying Independence via Joint Distributions

Source manuscript: chapter 6, page 10

Example (verifying independence)

① If $X, Y \sim \text{Unif}\{1, \ldots, 6\}$ are independent and $S = X + Y$, then $p_{X,S}(1, 12) = \mathbb{P}(X = 1,\ S = 12) = 0$, but

$$p_X(1)\, p_S(12) = \mathbb{P}(X = 1) \cdot \mathbb{P}(S = 12) = \frac{1}{6} \cdot \frac{1}{36} = \frac{1}{216} \neq 0,$$

so X, S are not independent.

② If $(X,Y) \sim \text{Unif}\Big(\boxed{}_{a}^{d}{}_{b}{}_{c} \Big)$, i.e. $(X,Y) \sim \text{Unif}\big([a,b]\times[c,d]\big)$, then

$$f_{X,Y}(x,y) = \frac{1}{(b-a)(d-c)}\,\mathbf{1}_{[a,b]\times[c,d]}(x,y) = \frac{1}{b-a}\,\mathbf{1}_{[a,b]}(x)\cdot\frac{1}{d-c}\,\mathbf{1}_{[c,d]}(y) \overset{\text{exercise}}{=} f_X(x)\cdot f_Y(y),$$

so X, Y are independent.

③ Suppose (X,Y) have joint density

$$f(x,y) = \begin{cases} \dfrac{7}{\sqrt{2\pi}}\exp\Big(-\dfrac{x^2}{2}-7y\Big) & \text{if } y>0, \\ 0 & \text{otherwise.}\end{cases}$$

Write

$$f(x,y) = \Big(7e^{-7y}\,\mathbf{1}_{[0,\infty)}(y)\Big)\cdot\frac{1}{\sqrt{2\pi}}\,e^{-x^2/2}.$$

These are both univariate densities, so X, Y are independent: indeed

$$\begin{aligned} f_X(x) = \int_{\mathbb{R}} f(x,y)\,dy &= \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\int_{\mathbb{R}} 7e^{-7y}\,\mathbf{1}_{[0,\infty)}(y)\,dy \\ &= \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\Big[-e^{-7y}\Big]_0^{\infty} = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}, \end{aligned}$$

and likewise $f_Y(y) = \int_{\mathbb{R}} f(x,y)\,dx = 7e^{-7y}\,\mathbf{1}_{[0,\infty)}(y)$; so $f = f_X\cdot f_Y$.

Source manuscript: chapter 6, page 11

④ Roll 2 dice repeatedly until the sum is either 2 or 6. Let $N :=$ # of rolls and $X :=$ the sum of the 2 dice on the N-th roll. Since

$$\mathbb{P}(\text{sum } 2) = \frac{1}{36}, \qquad \mathbb{P}(\text{sum } 6) = \frac{5}{36}, \qquad \text{so} \qquad \mathbb{P}(\text{sum 2 or 6}) = \frac{6}{36} = \frac{1}{6},$$

$N \sim \text{Geom}(\frac{1}{6})$, and

$$\mathbb{P}(N=n) = \Big(\frac{5}{6}\Big)^{n-1}\cdot\frac{1}{6} \qquad \text{and} \qquad \mathbb{P}(N=n,\ X=6) = \Big(\frac{5}{6}\Big)^{n-1}\cdot\frac{5}{36}.$$

Then

$$\mathbb{P}(X=6) = \sum_{n\geq 1}\mathbb{P}(N=n,\ X=6) = \frac{5}{36}\sum_{n\geq 1}\Big(\frac{5}{6}\Big)^{n-1} = \frac{5}{36}\cdot\frac{1}{1-\frac{5}{6}} = \frac{5}{6},$$

so $\mathbb{P}(N=n,\ X=6) = \mathbb{P}(N=n)\cdot\mathbb{P}(X=6)$. (Likewise, $\mathbb{P}(N=n,\ X=2) = \mathbb{P}(N=n)\cdot\mathbb{P}(X=2)$.) So N, X are independent.

□

Exercise. If $(Y_i)_{i\geq 1}$ are independent identically distributed random variables and $B \subseteq \mathbb{R}$ is such that $\mathbb{P}(Y_i \in B) > 0$, and $N = \inf\{n \geq 1 :\ Y_n \in B\}$, then N and Y_N are independent. (What are their distributions?)[1]

[1] With $p := \mathbb{P}(Y_1 \in B)$: $N \sim \text{Geom}(p)$, and Y_N has the distribution of Y_1 conditioned on $\{Y_1 \in B\}$, i.e. $\mathbb{P}(Y_N \in A) = \mathbb{P}(Y_1 \in A \mid Y_1 \in B)$. Example ④ is the case where Y_i is the sum of the i-th pair of dice and $B = \{2,6\}$ (there $\mathbb{P}(X=6) = \frac{5/36}{6/36}$); the urn of textbook Exercise 6.32 is another.

17.6 Independence of Discrete and Continuous Random Variables

Fact

Let D be a discrete random variable and C be a continuous random variable. Then D and C are independent $\iff$ for all $x \in S_D$ and all real $a < b$,

$$\mathbb{P}(D = x,\ a < C < b) = p_D(x) \cdot \int_a^b f_C(y)\, dy.$$

For $\vec{D} = (D_1, \ldots, D_n)$, $\vec{C} = (C_1, \ldots, C_m)$: replace x by $\vec{x}$, and $[a, b]$ by $[a_1, b_1] \times \cdots \times [a_m, b_m]$.

(As C is continuous, it makes no difference whether the interval is open or closed. In the vector version the factor $\int_a^b f_C$ becomes $\mathbb{P}\big(\vec{C} \in [a_1, b_1] \times \cdots \times [a_m, b_m]\big)$, which is the integral of the joint density of $\vec{C}$ over the box when $\vec{C}$ has one.)

Source manuscript: chapter 6, page 12

Example (cyclists and pedestrians)

Cars kill cyclists in Qc at a rate of one per 21 days. Rate → exponential: the number of days until the next death, C, is Exp(λ) with $\lambda = \frac{1}{21}$, with density $f_C(x) = \lambda e^{-\lambda x}\, \mathbf{1}_{[0,\infty)}(x)$. Independently, cars kill pedestrians in Qc at a rate of one per 7 days: the number of days until the next death, P, is Exp(μ) with $\mu = \frac{1}{7}$, with density $f_P(y) = \mu e^{-\mu y}\, \mathbf{1}_{[0,\infty)}(y)$.
Joint distribution: $f_{C,P}(x, y) = f_C(x) \cdot f_P(y) = \lambda\mu e^{-\lambda x - \mu y}$ for $x, y \geq 0$ (and 0 otherwise). So

$$\begin{aligned}\mathbb{P}(C < P) &= \iint_{0 \leq x < y} f_{C,P}(x, y)\, dy\, dx = \iint_{0 \leq x < y} \lambda\mu e^{-\lambda x - \mu y}\, dy\, dx \\ &= \int_0^\infty \lambda e^{-\lambda x} \Big(\int_x^\infty \mu e^{-\mu y}\, dy \Big) dx = \int_0^\infty \lambda e^{-\lambda x} e^{-\mu x}\, dx = \frac{\lambda}{\lambda + \mu}.\end{aligned}$$

Let $T = \min(C, P)$; then for $x \geq 0$

$$\mathbb{P}(T > x) = \mathbb{P}(C > x,\ P > x) = \mathbb{P}(C > x) \cdot \mathbb{P}(P > x) = e^{-\lambda x} e^{-\mu x} = e^{-(\lambda+\mu)x},$$

so $T \sim \text{Exp}(\lambda + \mu)$.

Lecture of November 3 ended here

Let

$$I = \mathbf{1}_{C<P} = \begin{cases} 1 & \text{if } C < P, \\ 0 & \text{otherwise} \end{cases} \sim \text{Ber}\Big(\frac{\lambda}{\lambda + \mu}\Big).$$

Then, for $0 \leq a < b$ (which is enough, as $T \geq 0$),

$$\begin{aligned}\mathbb{P}(I = 1,\ a \leq T \leq b) &= \mathbb{P}(a \leq C \leq b,\ P > C) = \int_a^b \int_x^\infty f_{C,P}(x, y)\, dy\, dx \\ &= \int_a^b \lambda e^{-\lambda x} \int_x^\infty \mu e^{-\mu y}\, dy\, dx = \int_a^b \lambda e^{-\lambda x - \mu x}\, dx \\ &= \frac{\lambda}{\lambda + \mu} \int_a^b (\lambda + \mu) e^{-(\lambda+\mu)x}\, dx \\ &= \mathbb{P}(I = 1) \cdot \int_a^b f_T(x)\, dx = \mathbb{P}(I = 1) \cdot \mathbb{P}(a \leq T \leq b).\end{aligned}$$

It follows that also

$$\begin{aligned}\mathbb{P}(I = 0,\ a \leq T \leq b) &= \mathbb{P}(a \leq T \leq b) - \mathbb{P}(I = 1,\ a \leq T \leq b) = \mathbb{P}(a \leq T \leq b) - \mathbb{P}(I = 1) \cdot \mathbb{P}(a \leq T \leq b) \\ &= \big(1 - \mathbb{P}(I = 1)\big)\, \mathbb{P}(a \leq T \leq b) = \mathbb{P}(I = 0)\, \mathbb{P}(a \leq T \leq b).\end{aligned}$$

So I, T are independent (by the fact above, with $D = I$, $C = T$). With the numbers: $\mathbb{P}(C < P) = (1/21)/(1/21 + 1/7) = 1/4$ and $\mathbb{E}T = 1/(\lambda + \mu) = 21/4$ days — the next death is a cyclist's with probability $\frac{1}{4}$, however soon or late it comes. □

17.7 The Joint PDF from the Joint CDF

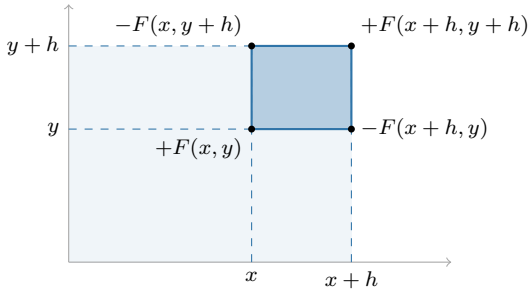
Source manuscript: chapter 6, page 13

Fact

If X, Y have joint CDF $F(x, y)$ and joint PDF $f(x, y)$, then

$$f(x, y) = \frac{\partial^2}{\partial x\, \partial y} F(x, y)$$

for all x, y such that $\dfrac{\partial^2}{\partial x\, \partial y} F(x, y)$ is defined and f is continuous at (x, y).



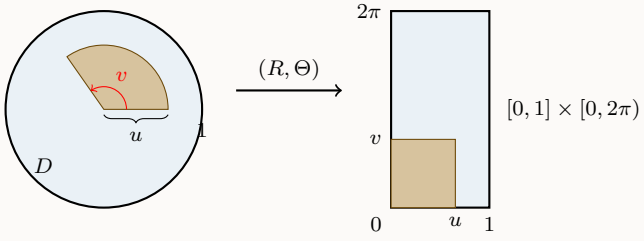
Each $F(\cdot, \cdot)$ is the probability of the quadrant to the lower left of its corner; with the signs shown, everything outside the small square cancels:

$$\begin{aligned}
\mathbb{P}\big((X - x) \in (0, h],\ (Y - y) \in (0, h]\big) &= F(x+h, y+h) - F(x, y+h) - F(x+h, y) + F(x, y) \\
&= \mathbb{P}(x < X \le x + h,\ y < Y \le y + h) \\
&= \big(F(x+h, y+h) - F(x+h, y)\big) - \big(F(x, y+h) - F(x, y)\big) \\
&\approx h \cdot \frac{\partial}{\partial y} F(x+h, y) - h \cdot \frac{\partial}{\partial y} F(x, y) \ \approx\ h \cdot h \cdot \frac{\partial}{\partial x}\Big(\frac{\partial}{\partial y} F(x, y)\Big),
\end{aligned}$$

while the same probability is $\int_x^{x+h} \int_y^{y+h} f(s, t)\, dt\, ds \approx h^2 f(x, y)$ when f is continuous at (x, y); comparing the two gives the fact.

Example (uniform point in the disk, in polar coordinates)

$(X, Y) \sim \text{Unif}(D)$, $D = B(\vec{0}, 1) = \{(x, y) \in \mathbb{R}^2 :\ \sqrt{x^2 + y^2} \le 1\}$, and let (R, Θ) be its polar coordinates, $R \in [0, 1]$, $\Theta \in [0, 2\pi)$. Joint CDF? Joint PDF?



For $0 \le u \le 1$ and $0 \le v \le 2\pi$, $\{R \le u,\ \Theta \le v\}$ is the event that (X, Y) lies in the sector of radius u and angle v, which has area $\frac{1}{2}u^2 v$; so

$$F_{R,\Theta}(u,v) = \mathbb{P}(R \le u,\ \Theta \le v) = \mathbb{P}\big((X,Y) \in \text{sector}\big) = \frac{\frac{1}{2}u^2 v}{\pi} = u^2 \cdot \frac{v}{2\pi},$$

and

$$f_{R,\Theta}(u,v) = \frac{\partial^2}{\partial u\,\partial v} F_{R,\Theta}(u,v) = \frac{u}{\pi} \qquad \text{for } 0 < u < 1,\ 0 < v < 2\pi \quad (0 \text{ otherwise}).$$

The marginal densities are

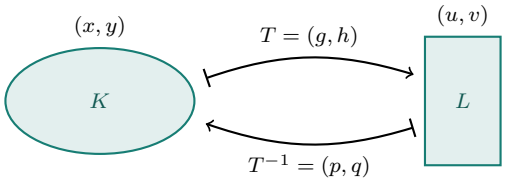
$$f_R(u) = \int_0^{2\pi} \frac{u}{\pi}\, dv = 2u \quad (0 < u < 1), \qquad f_\Theta(v) = \int_0^1 \frac{u}{\pi}\, du = \frac{1}{2\pi} \quad (0 < v < 2\pi).$$

So $f_{R,\Theta} \equiv f_R \cdot f_\Theta$, so R and Θ are independent. □

17.8 Transformation of Joint Densities

Source manuscript: chapter 6, page 14

Suppose (X, Y) has joint density f, and $(U, V) = T(X, Y)$, i.e. $U = g(X, Y)$, $V = h(X, Y)$. Can we find a joint density for U, V?



Assume there are regions $K, L \subseteq \mathbb{R}^2$ such that

- $f \equiv 0$ outside K;
- the function $(x, y) \overset{T}{\longmapsto} \big(g(x,y), h(x,y)\big)$ is a bijection from K to L.

Then we can define an inverse of T (restricted to L), $T^{-1} : (u, v) \longmapsto \big(p(u,v), q(u,v)\big)$, so that

$$x = p\big(g(x,y), h(x,y)\big), \qquad y = q\big(g(x,y), h(x,y)\big) \qquad \text{for } (x,y) \in K.$$

Assume $\frac{\partial p}{\partial u}, \frac{\partial p}{\partial v}, \frac{\partial q}{\partial u}, \frac{\partial q}{\partial v}$ all exist and are continuous on L, and that the matrix of partial derivatives

$$\text{Jac}(u,v) = \begin{bmatrix} \frac{\partial p}{\partial u}(u,v) & \frac{\partial p}{\partial v}(u,v) \\ \frac{\partial q}{\partial u}(u,v) & \frac{\partial q}{\partial v}(u,v) \end{bmatrix} \qquad \text{has} \qquad \det\big(\text{Jac}(u,v)\big) \neq 0 \quad \text{for } (u,v) \in L.$$

Theorem (transformation of joint densities)

Under these assumptions, the joint density of (U, V) is

$$f_{U,V}(u,v) = f_{X,Y}\big(p(u,v), q(u,v)\big) \cdot \big|\det\big(\text{Jac}(u,v)\big)\big| \qquad \text{for } (u,v) \in L,$$

and $f_{U,V}(u,v) = 0$ for $(u,v) \notin L$.

The factor $|\det(\text{Jac}(u,v))|$ is the derivative of the *inverse* map, which is the reciprocal of the derivative of the map: in one dimension it "is" $\dfrac{1}{g'(g^{-1}(y))}$, as in Fact 5.24.

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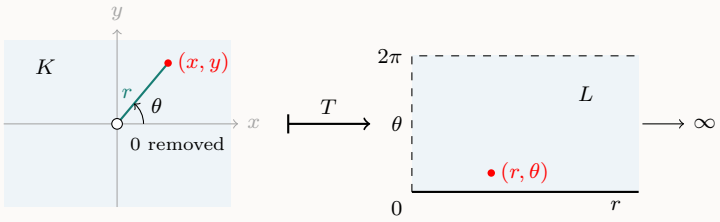
Example (independent standard normals in polar coordinates)

X, Y independent $\mathcal{N}(0,1) \longrightarrow$ joint density

$$f(x,y) = \varphi(x)\varphi(y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}.$$

$(X,Y) \overset{T}{\longmapsto} (R,\Theta)$, where $R = \sqrt{X^2+Y^2}$ and $\Theta \in [0,2\pi)$ is the polar angle of (X,Y), i.e. $(X,Y) = R(\cos\Theta, \sin\Theta)$; the inverse is $(r,\theta) \overset{T^{-1}}{\longmapsto} (x,y)$ with

$$x = p(r,\theta) = r\cos\theta, \qquad y = q(r,\theta) = r\sin\theta, \qquad K = \mathbb{R}^2 \setminus \{(0,0)\}, \qquad L = (0,\infty)\times[0,2\pi).$$



The partial derivatives are

$$\frac{\partial x}{\partial r} = \cos\theta, \qquad \frac{\partial x}{\partial \theta} = -r\sin\theta, \qquad \frac{\partial y}{\partial r} = \sin\theta, \qquad \frac{\partial y}{\partial \theta} = r\cos\theta,$$

so

$$\left|\det\big(\mathrm{Jac}(r,\theta)\big)\right| = \left|\cos\theta \cdot r\cos\theta - (-r\sin\theta)\sin\theta\right| = r\cos^2\theta + r\sin^2\theta = r.$$

Hence, for $(r,\theta) \in L$,

$$f_{R,\Theta}(r,\theta) = f\big(p(r,\theta), q(r,\theta)\big) \cdot \left|\det\big(\mathrm{Jac}(r,\theta)\big)\right| = \frac{1}{2\pi}\exp\Big(-\frac{1}{2}\big(r^2\cos^2\theta + r^2\sin^2\theta\big)\Big)\cdot r$$
$$= \frac{1}{2\pi}e^{-r^2/2}\cdot r = \frac{1}{2\pi}\cdot re^{-r^2/2} = \underbrace{f_\Theta(\theta)}_{\mathrm{Unif}[0,2\pi)} \cdot \underbrace{f_R(r)}_{\text{Rayleigh}} :$$

$\Theta \sim \mathrm{Unif}[0,2\pi)$, R is *Rayleigh*-distributed (density $re^{-r^2/2}$ on $(0,\infty)$), and R and Θ are independent. □

Exercise. If $U \sim \mathrm{Unif}[0,1]$ then $\sqrt{-2\ln U}$ is Rayleigh. So we can generate (X,Y) independent $\mathcal{N}(0,1)$ from U_1, U_2 independent $\mathrm{Unif}[0,1]$, by setting $\Theta = 2\pi U_1$, $R = \sqrt{-2\ln U_2}$, then $X = R\cos\Theta$, $Y = R\sin\Theta$.[2]

[2]This is the *Box–Muller* method. For the first claim: for $r \geq 0$, $\mathbb{P}\big(\sqrt{-2\ln U} > r\big) = \mathbb{P}\big(U < e^{-r^2/2}\big) = e^{-r^2/2} = \int_r^\infty s e^{-s^2/2}\,ds$, the Rayleigh tail.